

Title: Divergent CRAM formation and persistence across freshwater and marine environments revealed by AI-expanded MS/MS libraries

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Supplementary Information

Supplementary Note 1. Training set design and molecular representation

Rationale for the training-set design

A central challenge in generating environmentally realistic CRAM candidates is the extreme sparsity of strict CRAM structures in existing molecular databases. Although CRAM-like molecules are widely inferred from marine DOM measurements, only a very small fraction of molecules in large public databases simultaneously satisfy the structural constraints typically used to define CRAM, including multiple carboxyl groups, predominantly aliphatic carbon, moderate oxygenation, and at least one alicyclic ring. In the working dataset assembled for this study, only approximately 16 of 5,842 molecules met all four strict CRAM criteria, corresponding to about 0.3% of the training set. This sparsity is consistent with the observation that fewer than 500 structures in PubChem satisfy strict CRAM-like criteria despite the database containing more than 100 million compounds. These constraints make direct strict-CRAM-only training impractical and would likely cause a generative model to overfit a very narrow and unrepresentative subset of structures.

To address this limitation, we deliberately defined the training corpus as a CRAM and CRAM-like molecular space rather than a strict CRAM-only set. The objective was to expose the model not only to end-member CRAM architectures, but also to surrounding structural neighborhoods that plausibly correspond to CRAM precursors, intermediates, degradation products, or compositionally adjacent analogues. This broader training strategy was intended to improve generalization while preserving directionality toward CRAM-relevant chemical space through property conditioning during generation. In this sense, the training set was designed to reflect the realistic analytical challenge posed by DOM: environmental CRAM is not expected to emerge from a chemically isolated class, but from a continuum of oxidized, carboxyl-rich, and partially alicyclic precursor molecules.

Composition of the training set

The curated training set contained 5,842 molecules assembled from four complementary sources chosen to capture distinct structural features relevant to CRAM chemistry. First, 4,234 carboxylate-containing molecules were collected from the PubChem BioAssay database to enrich the dataset in oxygen-rich and carboxyl-bearing molecular frameworks. Second, 928 alicyclic structures were extracted from natural-product collections, including COCONUT and NPDB, to represent ring systems and scaffold motifs relevant to alicyclic CRAM architectures. Third, 664 dicarboxylic acids were drawn from synthetic polymer precursor libraries to broaden representation of multi-carboxylated aliphatic backbones. Finally, 16 hypothetical CRAM-like structures were included based on spectroscopic reconstructions of marine DOM, providing chemically informed exemplars of the target region of structural space.

This source composition was not intended to imply that each database directly represents marine DOM. Rather, the objective was to assemble a chemically diverse training corpus containing the core substructures, oxidation patterns, and functional-group combinations needed for the model to learn CRAM-relevant generative rules. The resulting set intentionally combined biologically derived molecules, synthetic analogues, and hypothesized DOM-like structures so that the model

could learn both realistic scaffold diversity and environmentally meaningful functional-group organization.

Molecular standardization and string representation

All molecules were standardized prior to model training using RDKit-based preprocessing. Duplicate entries were removed, tautomers were canonicalized where appropriate, and molecules were neutralized when possible under the standardization workflow. Canonical SMILES strings were generated for storage, indexing, and structure comparison. SELFIES representations were then derived from the standardized molecules and used as the output target for the decoder. SELFIES was selected because it provides a robust string-based molecular representation in which syntactic validity is guaranteed by construction, thereby reducing the prevalence of invalid outputs during autoregressive decoding compared with conventional SMILES-based generation.

The use of SELFIES as the decoder target should be understood as a practical choice for robust generation rather than as a replacement for graph-level molecular encoding. In this study, graph representations were used to encode structural information into latent space, whereas SELFIES served as a chemically constrained sequence representation for molecular reconstruction and generation. This combined graph-to-sequence design allowed the model to leverage both explicit structural topology during encoding and robust string decoding during generation.

Definition of strict CRAM compliance

Strict CRAM compliance was evaluated independently from training-set inclusion. A molecule was considered CRAM-compliant if it simultaneously satisfied four criteria: at least two carboxyl groups, at least 70% aliphatic carbon, an O/C ratio between 0.4 and 0.6, and the presence of at least one alicyclic ring excluding aromatic rings. These criteria were used for post hoc assessment of generated molecules and not as a hard filter for assembling the training set. This distinction is important. If strict CRAM compliance had been used as the sole inclusion rule at the training stage, the available molecular diversity would have been too limited to support meaningful generative learning. By decoupling training-set inclusion from strict CRAM evaluation, the model could learn a broader chemically relevant space while still being judged against the more restrictive CRAM definition during downstream validation.

Graph-based molecular representation

For graph encoding, each molecule was represented as an atom–bond graph in which atoms were treated as nodes and covalent bonds as edges. Each atom was described by a 163-dimensional feature vector grouped into eight conceptually distinct categories. These were designed to capture not only standard structural attributes used in graph neural networks, but also chemical patterns that were hypothesized to be relevant to CRAM-like behavior in aquatic environments.

The first category comprised basic atomic and topological features and included atom type, hybridization, formal charge, ring-size membership, and core physicochemical descriptors such as valence, aromaticity, atomic mass, electronegativity, van der Waals radius, ionization energy, electron affinity, covalent radius, Gasteiger partial charge, and local connectivity features. These descriptors provide the baseline chemical information required to distinguish common atom environments and local bonding patterns.

The second category captured ring-chain structural features, including ring size, normalized ring count, bridgehead atom flags, chain length for non-ring atoms, and alicyclic versus aromatic ring classification. These descriptors were included because the balance between cyclic and chain-like organization is central to the operational definition of CRAM and to its expected fragmentation behavior.

The third category encoded stereochemical features, including tetrahedral chirality, stereochemical specification flags, and neighbor diversity around stereogenic centers. Although stereochemistry is not explicitly resolved in formula-level DOM analysis, it contributes to scaffold diversity and may affect the structural realism of generated molecules.

The fourth category comprised marine DOM-related features that were designed to reflect environmentally relevant structural motifs. These included descriptors for carboxylate positioning, alicyclic carbon patterns, hydroxyl and carbonyl occurrences, and redox-active motifs such as quinone-like and phenolic functionalities. These features were not intended as definitive markers of marine DOM identity, but rather as heuristics to guide the model toward chemically plausible regions of DOM-relevant structural space.

The fifth category contained synthetic accessibility-related features, including bridgehead carbons, spiro centers, fused ring systems, strained ring motifs, and atom-level connectivity patterns relevant to molecular complexity. These features were included to help the model distinguish realistic scaffold topologies from graph-theoretically valid but chemically implausible constructions.

The sixth category encoded terpenoid pattern features, such as SMARTS-based matches to isoprene-derived motifs, cyclohexane and cyclopentane ring patterns, and methyl branching arrangements characteristic of terpenoid-like precursor architectures. These descriptors were included because terpenoid-related structures were hypothesized to contribute to marine CRAM formation and later emerged as a dominant biosynthetic signature in the environmental network analysis.

The seventh category described functional-group features, including carboxylic acid, carboxylate, hydroxyl, amine, carbonyl, ester, amide, sulfone, and phosphate motifs. These descriptors capture the oxygen-rich and heteroatom-rich functionalities expected to influence both CRAM compliance and environmental plausibility.

The eighth category encoded oxidation-state-related features, including carbon oxidation proxies such as oxygen-neighbor count and carbonyl-bond presence, together with positional encoding for heteroatoms such as O, N, and S. These features were included because CRAM formation in the main analysis was hypothesized to be strongly linked to oxidative processing, and thus oxidation-state-aware encoding was expected to improve navigation of relevant chemical space.

Design principles behind the feature set

The feature set was designed under three principles. First, it had to preserve sufficient general chemical information for stable graph-based learning. Second, it had to enrich the representation with motifs plausibly relevant to CRAM chemistry, including carboxylation state, alicyclic scaffold organization, and oxidation-related functionality. Third, it had to remain broad enough to avoid hard-coding a single preconceived CRAM structure into the model. The resulting

representation should therefore be viewed not as a definitive descriptor system for marine DOM, but as a chemically informed encoding strategy intended to improve generative access to an undersampled region of environmentally relevant structural space.

Role of Supplementary Note 1 within the overall workflow

This training-set and molecular-representation design defines the chemical search space from which the downstream generative model operates. Because strict CRAM structures are too sparse to support direct training, the broader CRAM-like corpus serves as a surrogate learning space, while property conditioning and downstream environmental filtering constrain generation toward plausible CRAM candidates. The quality of this initial design is therefore foundational to all later analyses, including candidate annotation, reaction-network reconstruction, and biosynthetic provenance analysis. In this study, the aim was not to claim that the training set fully represents environmental CRAM, but rather to construct a chemically grounded starting space from which environmentally plausible and analytically useful CRAM candidates could be generated at scale.

Supplementary Note 2. Model architecture, training, and ablation analysis

Overview of the model design

To generate environmentally plausible CRAM candidates from a chemically sparse training space, we designed a graph-encoder Transformer variational autoencoder (DeepCRAMVAE backbone) with an Enhanced Hierarchical GAT encoder fused to a RingChainEncoder, a 64-dimensional variational latent code, a 17-dimensional property head on z , and a causal Transformer SELFIES decoder. Despite the Property Conditioned TransformerVAE name, y is supervised by MSE on $\text{MLPprop}(z)$ rather than concatenated into the decoder. The selected production checkpoint is Property Conditioned TransformerVAE / ff2048.

The model was not designed as a general-purpose molecular generator. Instead, it was tailored to a specific environmental chemistry problem in which the target molecules are rare in public databases and must satisfy multiple interdependent constraints, including high carboxyl content, aliphatic-rich frameworks, moderate oxidation state, and alicyclic character under environmentally plausible aqueous conditions. This motivation distinguishes the present setting from the more common pharmaceutical applications of molecular generation, where the objective is typically to optimize drug-likeness, bioactivity, or synthetic accessibility.

Graph attention encoder

The encoder processed each molecule as an atom–bond graph. Each atom is a 163-dimensional feature vector (eight chemically motivated blocks: $45+28+15+20+12+10+13+20$), linearly projected to $H = 256$, then an Enhanced Hierarchical GAT with four residual GAT layers (four attention heads) and multi-scale readout, fused with a Ring Chain Encoder that pools alicyclic/chain/junction/carboxyl masks via cross-attention. Dropout 0.1 is applied in the selected winner.

The fused graph embedding h in $\mathbb{R}^{256}$ parameterizes μ and $\log \sigma^2$ of a 64-dimensional latent Gaussian through MLPs, with $\log \sigma^2$ clamped to $[-10, 2]$. Latent samples use the reparameterization trick. Implemented encoder z is 64. The Multimodal Transformer VAE / Transformer VAE sweep arms use $z = 128$; the production winner uses $z = 64$.

Property-conditioning module

A central objective of the architecture was to direct generation toward CRAM-relevant chemical space without forcing the training set itself to be restricted to strict CRAM molecules. To achieve this, the sampled latent code was conditioned on five target properties: carboxyl-group count, ring classification, aliphatic carbon percentage, molecular weight, and benzene-ring presence. These properties were selected because together they define the most important structural axes that separate strict CRAM-like molecules from nearby but chemically distinct molecular classes. Carboxyl-group count and aliphatic character encode the dominant CRAM-defining compositional tendencies, ring classification captures the distinction between alicyclic and aromatic scaffold organization, molecular weight helps constrain the scale of generated molecules, and benzene presence serves as a practical handle on aromaticity.

Property supervision uses a molecule-level vector y in $\mathbb{R}^{17}$ predicted by a shared head $y = \text{MLP}_{\text{prop}}(z)$ with MSE loss (weight 0.5). The 17 dimensions cover size/shape, polarity/H-bonding, skeleton/rings, oxygenation, and marine/Hertkorn coordinates (O/C, H/C, DBE); slots 15–16 are unused zeros. This is latent property supervision, property head on z rather than classical CVAE concatenation of c into the decoder. The purpose is to keep z informative for Hertkorn placement and overall molecular realism rather than to post-filter after decoding.

Transformer SELFIES decoder

Latent z is mapped to a single memory token $\text{Linear}(z) \rightarrow \mathbb{R}^{H_{\text{dec}}}$. A causal Transformer decoder (eight heads) reconstructs SELFIES under teacher forcing; at inference it runs autoregressively from $\langle \text{SOS} \rangle$ until $\langle \text{EOS} \rangle$. The selected winner uses six decoder layers, feed-forward width 2048, $H = 256$, and dropout 0.1. Sequences are padded/truncated to $L_{\text{max}} = 150$. Vocabulary size is 53 (including $\langle \text{PAD} \rangle$, $\langle \text{SOS} \rangle$, $\langle \text{EOS} \rangle$), as deposited with the checkpoint.

SELFIES was selected as the decoder representation because it guarantees syntactic validity by construction, substantially reducing the frequency of invalid outputs relative to conventional SMILES-based decoding. In the present application, this was particularly important because the model was used to generate molecules at very large scale. A robust decoder was therefore necessary not only for reconstruction accuracy during training, but also for ensuring high validity during large-scale sampling of candidate structures.

Training objective

The training objective is $L = L_{\text{recon}} + \beta L_{\text{KL}} + 0.1 L_{\text{contr}} + 0.5 L_{\text{prop}}$, where L_{recon} is token cross-entropy (ignore pad), L_{KL} is the standard Gaussian KL, L_{contr} is supervised contrastive loss on z using CRAM/non-CRAM labels, and $L_{\text{prop}} = \text{MSE}(y_{\text{hat}}, y)$. Legacy config fields γ_{TC} and $\lambda_{\text{cram reward}}$ are not terms in this written loss. The total loss can be written as

The KL weight β follows cyclic annealing over four cycles in $[5 \times 10^{-5}, 3 \times 10^{-3}]$. Contrastive and property weights are fixed at 0.1 and 0.5 as in Methods.

The contrastive term on CRAM/non-CRAM labels and the 17D property head are auxiliary inductive biases on z , not hard classifiers imposed at decode time. They encourage latent structure that remains informative for Hertkorn-relevant coordinates while the Transformer decoder reconstructs SELFIES.

Training procedure

The curated training set comprises $n = 5,842$ molecules (Supplementary Note 1; $4,234 + 928 + 664 + 16$; 16 strict-CRAM exemplars 0.3%), divided 85%/15% for training/validation. Optimization used AdamW (10^{-4} , weight decay 10^{-6}), cosine LR annealing (floor 5×10^{-7}), gradient accumulation of 4, mixed precision (AMP), and gradient clipping at 1.0. Each sweep run trained for 100 epochs; the checkpoint with lowest validation loss was retained (winner val loss 0.7381). Enrichment $R = (5,760/10,000)/(16/5,842)$ uses this curated denominator.

Hyperparameter selection

Implemented production latent dimensionality is $z = 64$ (PropertyConditionedTransformerVAE / ff2048). Sweep arms with $z = 128$ (MultimodalTransformerVAE / TransformerVAE) are competitive but not selected; the ranking shows a deep, wide decoder matters more than enlarging z alone. Table 1 57.6% remains the $n = 10,000$ generation-rate sample and is not reassigned to a latent width. These hyperparameter choices should therefore be viewed not as universally optimal settings for environmental molecular generation, but as problem-specific values that stabilized training while promoting access to the desired CRAM-relevant region of chemical space. The main conclusion from the tuning stage was that model guidance toward CRAM cannot be achieved by latent sampling alone; explicit property conditioning is necessary.

Large-scale generation and initial performance

For large-scale candidate generation, 10 million latent vectors were sampled from the prior distribution, conditioned on target property values, and decoded into SELFIES strings. Candidate molecules were retained only if they passed RDKit sanitization and valence checks. Table 1 ($n = 10,000$) is the only tabulated generation-rate sample ($57.6\% \pm 1.6\%$ strict CRAM; $70.0\% \pm 1.5\%$ environmentally plausible of all 10,000), labelled as single-run binomial intervals; seed n is not recovered. The data product is 5.04 million environmentally plausible CRAM candidates. The $57.6\% / \sim 0.3\%$ 192 ratio compares conditioned generation with sparse training labels, not independent discovery. These values demonstrate that the model did not merely reproduce fragments of the training distribution but instead generated a large and chemically coherent candidate space strongly enriched in strict CRAM-like molecules. Because the training corpus itself contained very few strict CRAM examples, the observed enrichment further supports the view that the model learned a transferable representation of CRAM-relevant structural logic rather than memorizing rare end-member structures.

Baseline model comparisons

To contextualize model performance, we implemented four established generative baselines and trained them on the same 5,842-molecule dataset under matched train/validation splits and comparable evaluation settings: CharRNN, JT-VAE, GraphVAE, and MolGAN. CharRNN used a character-level LSTM decoder operating on SMILES strings, JT-VAE used fragment-based tree-structured decoding, GraphVAE generated graphs directly through node and edge prediction, and MolGAN used graph-based adversarial generation with reinforcement-learning-style property optimization. None of these baselines included the explicit property-conditioning module used in the present model.

This comparison should therefore be interpreted with some care. Performance differences among models reflect both architectural differences and the presence or absence of direct conditioning toward CRAM-relevant target properties. Even with this caveat, the comparison was informative because it provided a realistic reference frame for how well standard molecular generators perform in a chemically specialized environmental setting. Among the tested baselines, MolGAN performed best, but still achieved substantially lower strict CRAM compliance than the PropertyConditionedTransformerVAE / ff2048 winner. In the main analysis, the best baseline reached 27.6% strict CRAM compliance, whereas the full conditioned model achieved 57.6%.

Ablation analysis

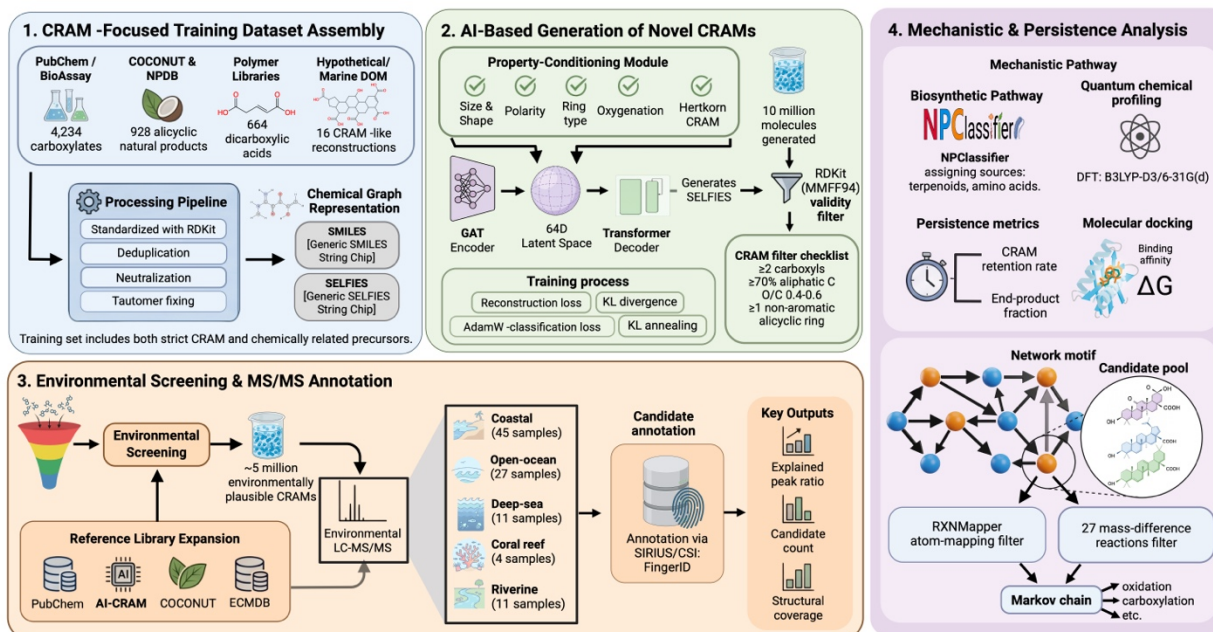
To isolate the contribution of individual design choices, we performed systematic ablation across 12 controlled model variants. These variants were used to test the impact of the major architectural components, including property-head supervision, graph-based encoding, SELFIES decoding, and Transformer decoder capacity. The eight configurations selected a deep, wide Transformer decoder (six layers, FF 2048) at $z = 64$ over enlarging the latent alone.

The advantage of the full model does not arise solely from using a more expressive neural network, but from explicitly aligning the generative process with chemically meaningful target properties. In other words, access to CRAM-like space was not recovered simply by better latent compression or more stable decoding. It depended primarily on a conditioning strategy that translated the working chemical definition of CRAM into a generative bias. The remaining architectural choices then enhanced the model's ability to realize that bias efficiently and stably at large scale.

Interpretation of the model design

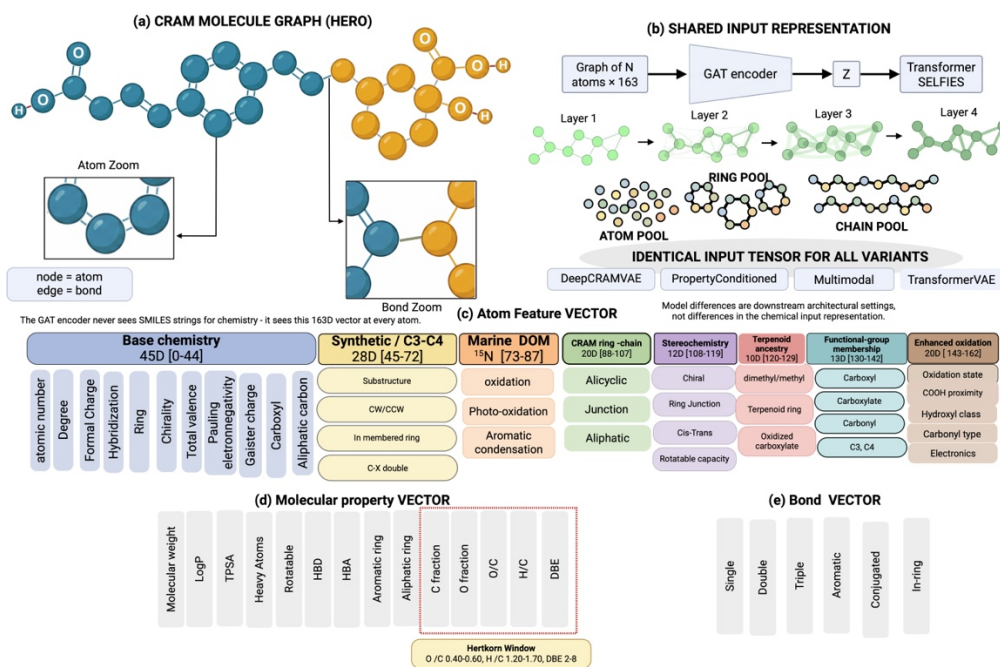
Taken together, the architecture, training protocol, and ablation results support a consistent interpretation of the model's role in this study. The graph encoder supplied chemically informed structural compression, the property-conditioning module directed exploration toward a sparse but environmentally relevant target region, the SELFIES decoder ensured robust large-scale generation, and the auxiliary CRAM-awareness term encouraged retention of the structural attributes needed for downstream compliance. The resulting model should therefore be understood less as a general-purpose molecular generator and more as a directed chemical-space expansion framework designed for environmentally constrained structural discovery.

Within the broader analytical workflow, this model provides the structural density required for downstream MS/MS matching and pathway reconstruction. Its value therefore lies not only in generative performance itself, but in enabling subsequent annotation-supported inference of environmentally divergent CRAM formation pathways.



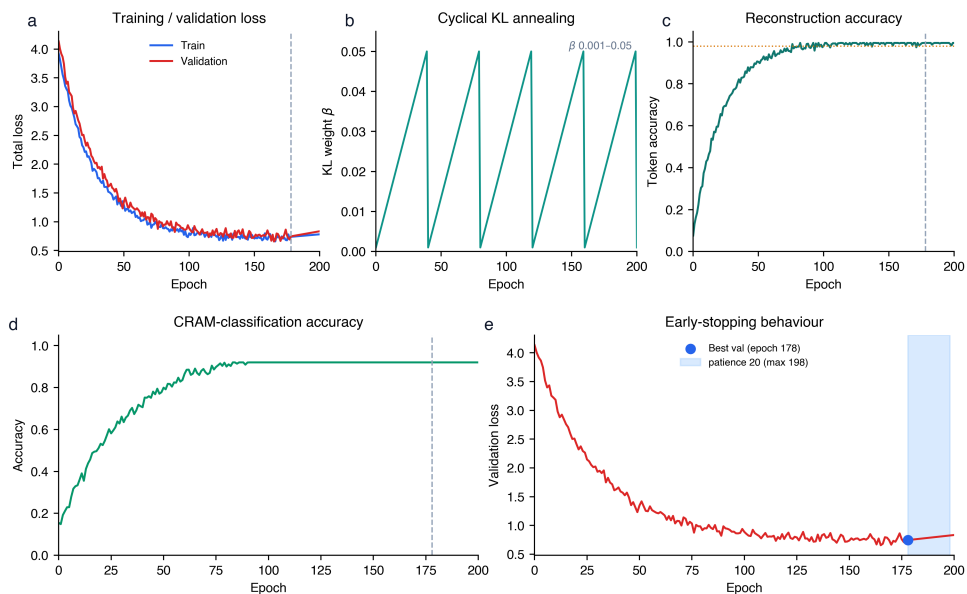
Supplementary Figure S1 — Overview study

(1) CRAM database; (2) AI for CRAM generation; (3) Environmental Screening and MS/MS annotation; (3) Mechanistic and Persistence Analysis



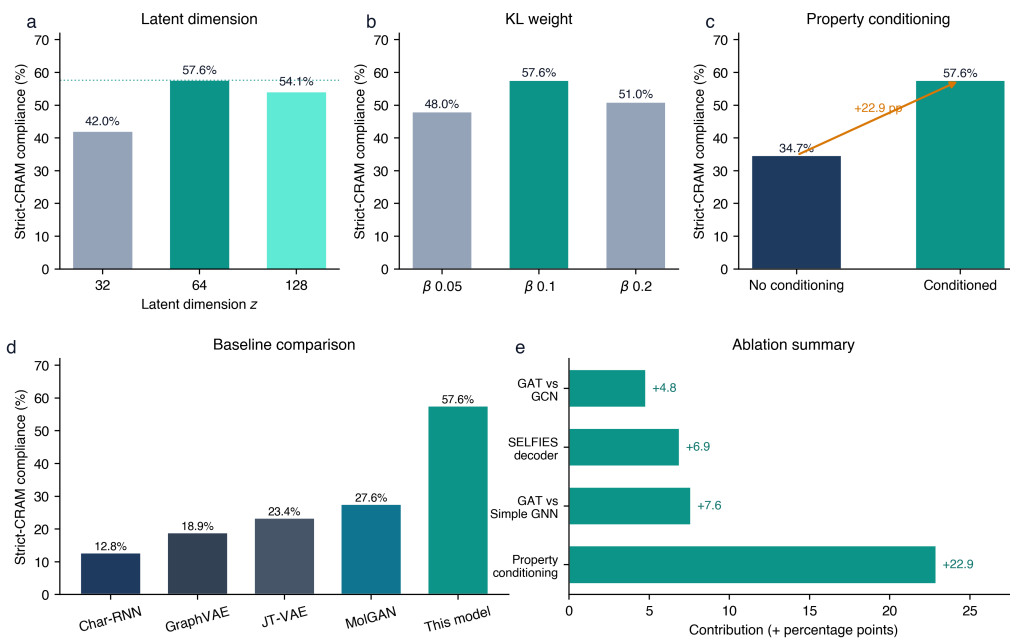
Supplementary Figure S2 — Property Conditioned TransformerVAE Architecture

(a) Explanation of Graph CRAM molecule; (b) Shared input representation of model; (c) Atom features; (d) Molecular property vector; (e) Bond vector



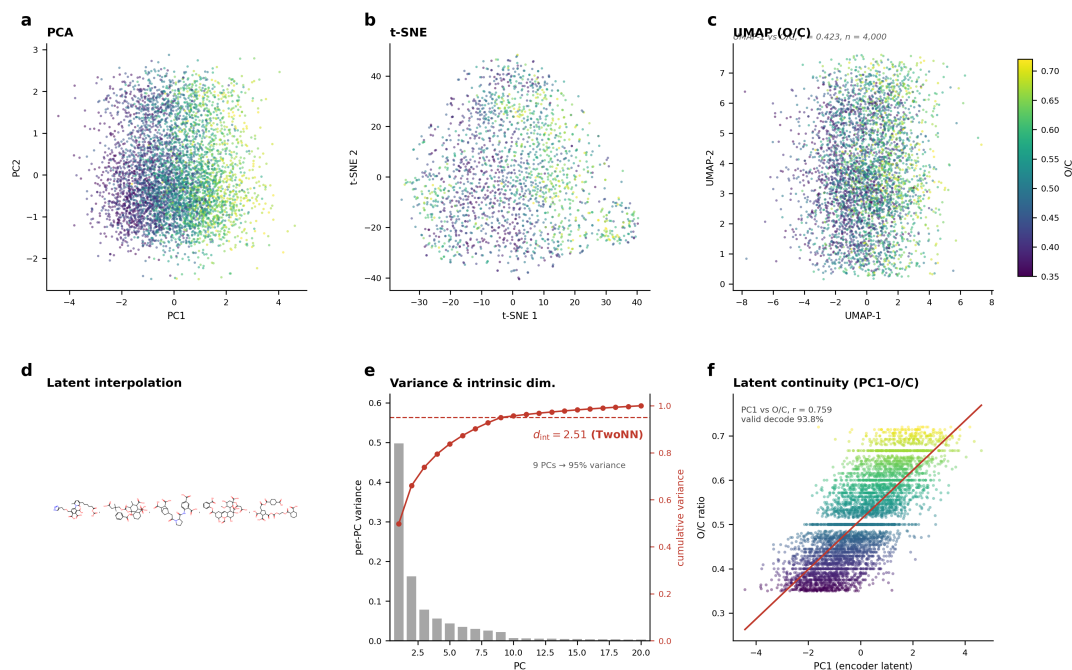
Supplementary Figure S3 — Model Training and Convergence

(a) loss curves; (b) KL-divergence; (c) reconstruction accuracy; (d) CRAM-classification accuracy; (e) early stopping



Supplementary Figure S4 — Hyperparameter Optimization and Ablation Analysis

(a) latent dimension effect; (b) KL weight effect; (c) property conditioning effect; (d) baseline comparison; (e) ablation summary



Supplementary Figure S5 — Latent-Space Validation

(a) PCA; (b) t-SNE; (c) UMAP; (d) latent interpolation; (e) TwoNN intrinsic dimensionality; (f) latent continuity.

Supplementary Note 3. Generated library quality, novelty, and environmental plausibility

Overview

The objective of the generative stage was not merely to maximize the number of valid molecules, but to construct a large candidate library that is simultaneously CRAM-enriched, structurally diverse, compositionally realistic, and environmentally plausible for aqueous aquatic systems. We therefore evaluated the generated library along four complementary axes: chemical validity and CRAM compliance, structural novelty and diversity, compositional agreement with experimentally observed marine DOM, and physicochemical plausibility under seawater-relevant conditions. Together, these analyses were used to determine whether the generated library could serve as a meaningful reference space for downstream MS/MS annotation and pathway reconstruction rather than as a large but chemically uninformative collection of molecules.

Chemical validity and strict CRAM compliance

Large-scale generation sampled 10 million latent vectors. The downstream data product is 5.04 million environmentally plausible CRAM structures. Table 1 ($n = 10,000$: 57.6% strict CRAM; 70.0% environmentally plausible) is the only audited generation-rate sample. A deposited companion production-funnel CSV exists, but production-scale percentages (including any 72.3% / 7.23M labels) are not audited from generation logs and are not live rates; the live data product remains 5.04 million Env-plausible structures. Arithmetic $10,000,000 \times 0.576 \times 0.700 = 4.03$ million, which is not 5.04 million; applying Table 1 rates to 10 million does not reconcile the two integers. Relative to the training set (16/5,842 strict CRAM), $R = (5,760/10,000)/(16/5,842) =$

210.3 [128.9, 343.2] compares conditioned generation with sparse labels rather than independent discovery. MolGAN reached 27.6% strict CRAM compliance without the property-conditioning module, so architectural superiority is not isolated from conditioning.

The representative 10,000-molecule validation subset further showed that the four individual CRAM-defining features were each frequently achieved at high rates, with 78.4% of molecules containing at least two carboxyl groups, 81.6% meeting the aliphatic-carbon threshold, 79.2% falling within the target O/C range, and 73.8% containing at least one alicyclic ring. Because the fully compliant fraction arises from the intersection of all four criteria, the 57.6% strict CRAM rate reflects not a single dominant filter but simultaneous satisfaction of multiple interdependent structural constraints. This is important because it indicates that the model is learning coordinated CRAM-like structural organization rather than independently overproducing one favorable feature, such as carboxylation alone.

Structural novelty and diversity

To assess whether the library represented genuine expansion of CRAM-like space rather than near-duplication of training molecules, novelty and diversity were evaluated using fingerprint-based analyses. Structural diversity was quantified using circular Morgan fingerprints (radius = 2, 2048 bits), pairwise Tanimoto similarity, and Murcko scaffold decomposition. Novelty was assessed by comparing each generated structure to the training set and classifying molecules as novel when their Tanimoto similarity to every training structure was below 0.9. This definition was selected to identify molecules that remain chemically close enough to the learned training manifold to be plausible, while excluding near-replicates of known structures. Latent-space organization was visualized using t-SNE applied to the learned latent representations to examine whether generated molecules occupied coherent but expanded neighborhoods of structural space.

Among the environmentally plausible strict-CRAM candidates retained for downstream analysis, the library contained 3.95 million unique Murcko scaffolds and exhibited a scaffold entropy of 21.8 bits, corresponding to 99.5% of the theoretical maximum under the observed scaffold count. This result indicates that the expanded library is not dominated by repeated decoration of a small number of backbone types but instead spans a broad architectural space of CRAM-like frameworks. In practical terms, this matters because pathway-level annotation and network reconstruction require multiple plausible candidate architectures per formula rather than a narrow family of near-isomeric variants. The high scaffold count therefore supports the interpretation that the library expands structural coverage in a chemically meaningful way.

Compositional realism relative to marine DOM

High novelty and diversity are valuable only if the generated structures remain anchored to environmentally relevant chemical space. We therefore compared the generated library to experimentally characterized marine DOM using major formula-derived compositional axes. Kolmogorov–Smirnov D statistics versus marine DOM (O/C 0.08, H/C 0.11, NOSC 0.09) are descriptive only; This compositional realism is an important complement to novelty. A generator can in principle produce highly diverse molecules that are nevertheless chemically irrelevant to natural DOM. Scaffold counts (3.95 million unique Murcko scaffolds) are a library-composition statistic.

Environmental plausibility screening

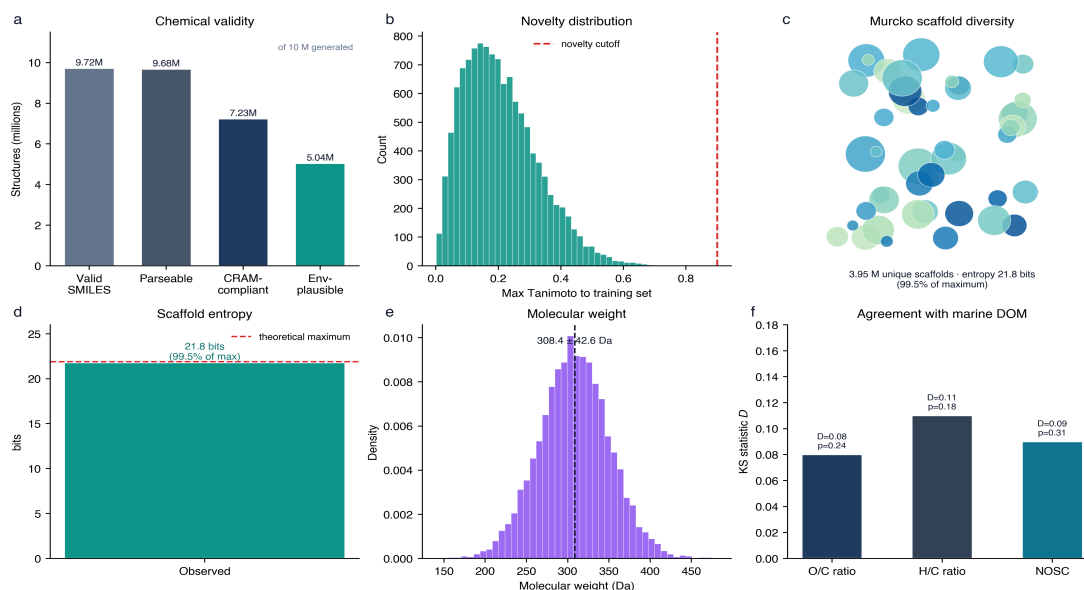
Because the central objective of this study was to analyze CRAM in dissolved environmental organic matter, an additional filtering stage was applied after chemical validity and strict CRAM assessment. Environmental plausibility was evaluated using two predicted physicochemical properties: aqueous solubility and lipophilicity under seawater-relevant conditions. Solubility was estimated with the ESOL model implemented in RDKit, and lipophilicity was evaluated using the distribution coefficient at pH 8.1, logD8.1, predicted with SwissADME. Molecules were flagged as environmentally implausible if their predicted solubility was below 0.1 mgmL⁻¹ or if logD8.1 exceeded 8. These thresholds were chosen to exclude molecules likely to partition strongly to particles, aggregate by hydrophobic collapse, or otherwise behave in ways inconsistent with dissolved organic matter in the water column.

The logic of this screening step is especially relevant for CRAM. Molecules with high carboxyl content are expected to retain substantial aqueous affinity through ionization and hydrogen bonding, whereas highly hydrophobic structures are less likely to persist as dissolved DOM even if they satisfy formal CRAM criteria. The plausibility filter was therefore applied not as a claim about absolute environmental existence, but as a conservative prioritization step to favor structures that are chemically compatible with dissolved-phase persistence in aquatic systems. This distinction is important: the filter does not prove that retained molecules occur in nature, but it reduces the likelihood that the downstream library is populated by formally valid yet environmentally implausible candidates.

Interpretation of library quality in the context of downstream analysis

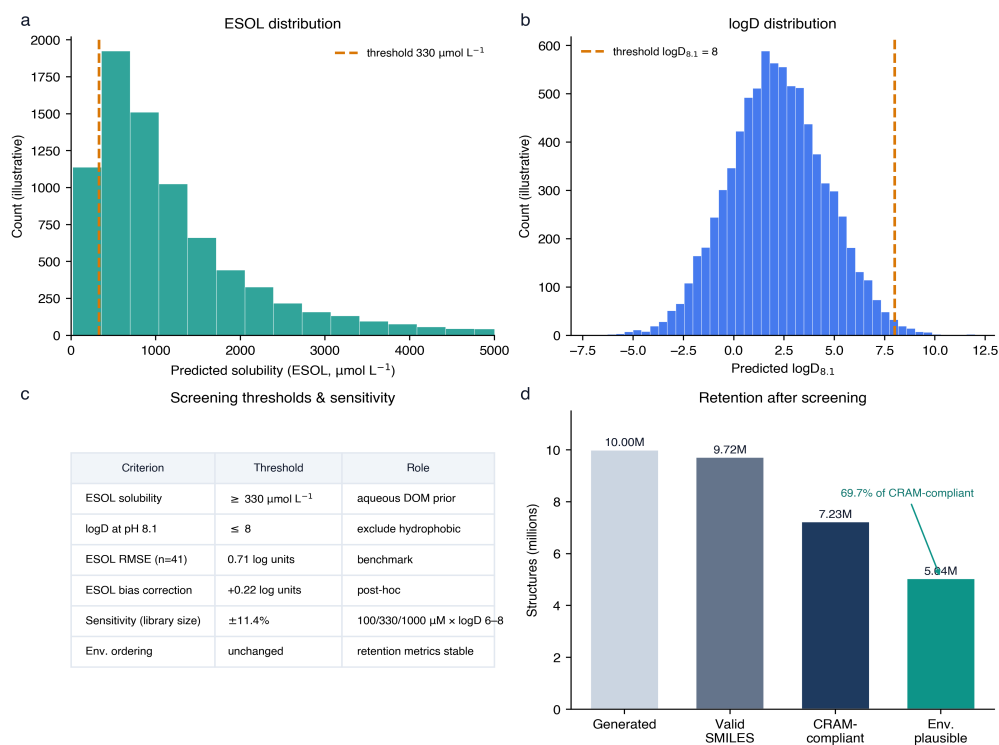
Taken together, the library-quality analyses support three conclusions. First, the generative model produced chemically stable outputs at scale and strongly enriched strict CRAM-like molecules relative to both the training set and standard baseline generators. Second, the resulting library was structurally broad, with very high scaffold diversity and substantial novelty relative to known molecules. Third, this diversity remained anchored to environmentally relevant chemical space through agreement with marine-DOM compositional descriptors and by explicit filtering against implausible solubility and lipophilicity profiles. The expanded library should therefore be interpreted not simply as a large collection of hypothetical structures, but as a chemically constrained candidate space suitable for annotation-supported inference in ultracomplex DOM mixtures.

Within the overall study workflow, these properties justify using the generated library as a reference layer for MS/MS matching and downstream pathway analysis. The role of Supplementary Note 3 is therefore to show that the library is large enough to expand coverage, diverse enough to increase structural alternatives, and chemically plausible enough to remain relevant to aquatic DOM rather than becoming an arbitrary exercise in molecular generation.



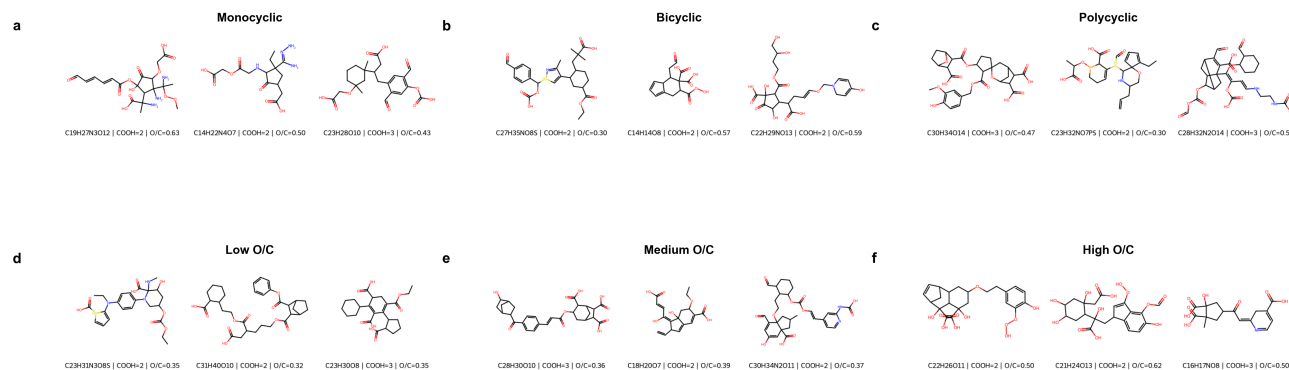
Supplementary Figure S6 — Chemical Validity, Novelty, and Scaffold Diversity

(a) validity stats; (b) novelty distribution; (c) scaffold diversity; (d) scaffold entropy; (e) MW distribution; (f) comparison with experimental analogues



Supplementary Figure S7 — Environmental Plausibility Screening

(a) ESOL distribution; (b) $\log D$ distribution; (c) screening thresholds; (d) sensitivity analysis; (e) retained structures; (f) before/after filtering



Supplementary Figure S8 — Representative AI-Generated CRAM Structures

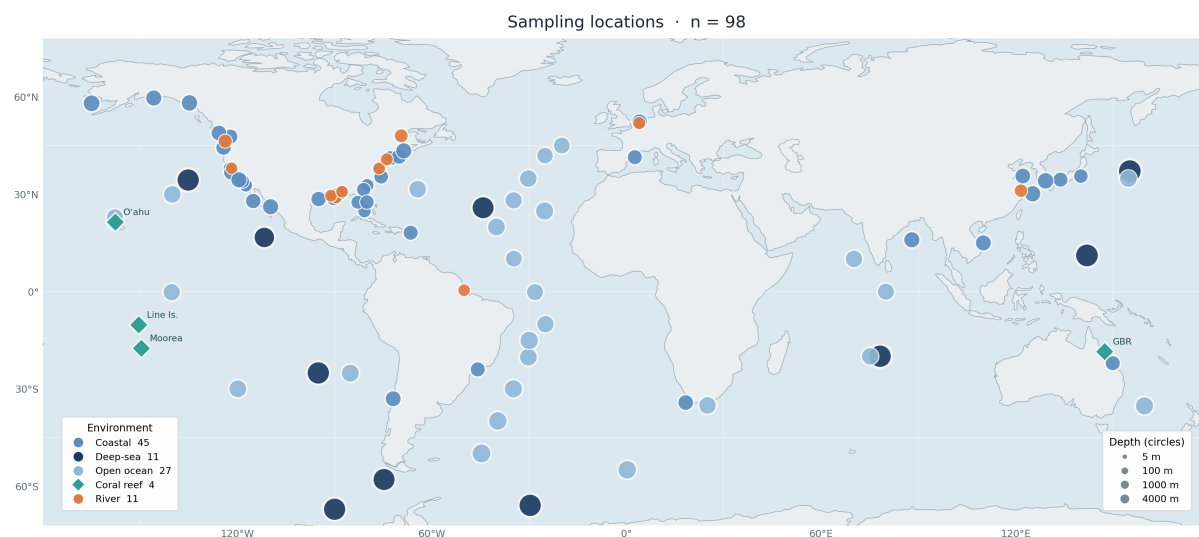
(a) monocyclic; (b) bicyclic; (c) polycyclic; (d) low O/C; (e) medium O/C; (f) highly oxidized

Supplementary Note 4. Environmental LC–MS/MS datasets and annotation workflow

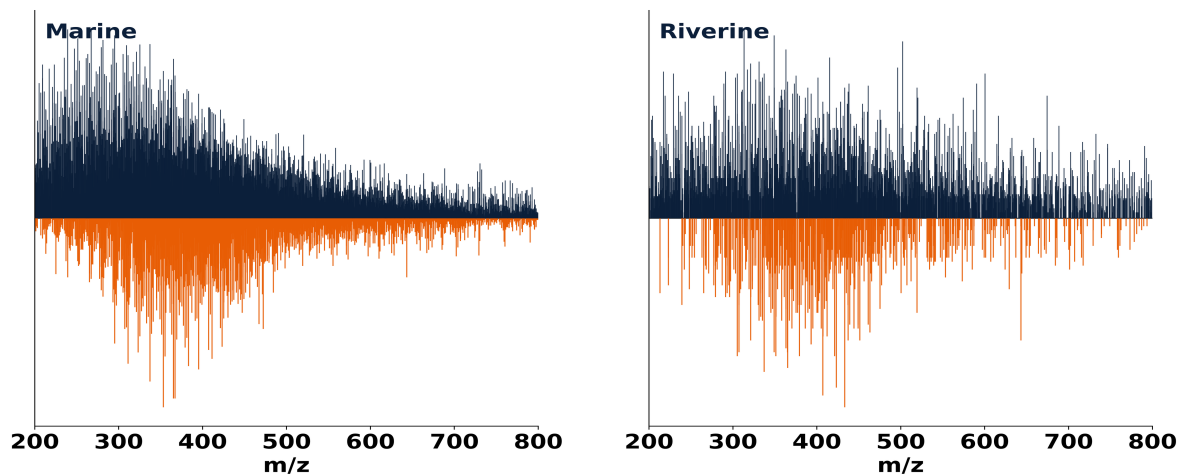
Overview

To evaluate whether the expanded CRAM library improves structural interpretation of environmental DOM, we integrated the generated candidate set with high-resolution LC–MS/MS datasets spanning multiple aquatic environments and compared annotation performance across alternative reference-library scenarios. The goal of this workflow was not only to increase the number of candidate annotations, but also to test whether CRAM-enriched structural coverage improves structural plausibility in the CRAM compositional region and thereby enables downstream pathway analysis. In the main manuscript, this workflow underpins the transition from formula-level characterization to annotation-supported inference of environmentally divergent CRAM formation pathways.

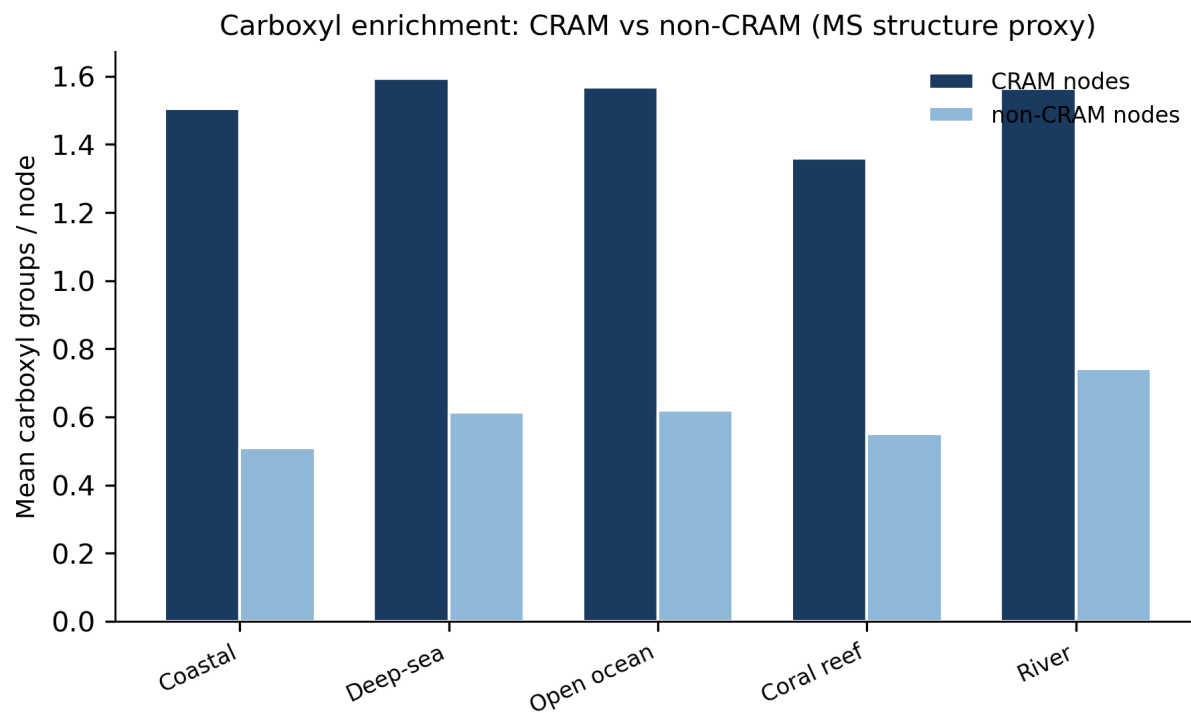
Environmental LC–MS/MS datasets



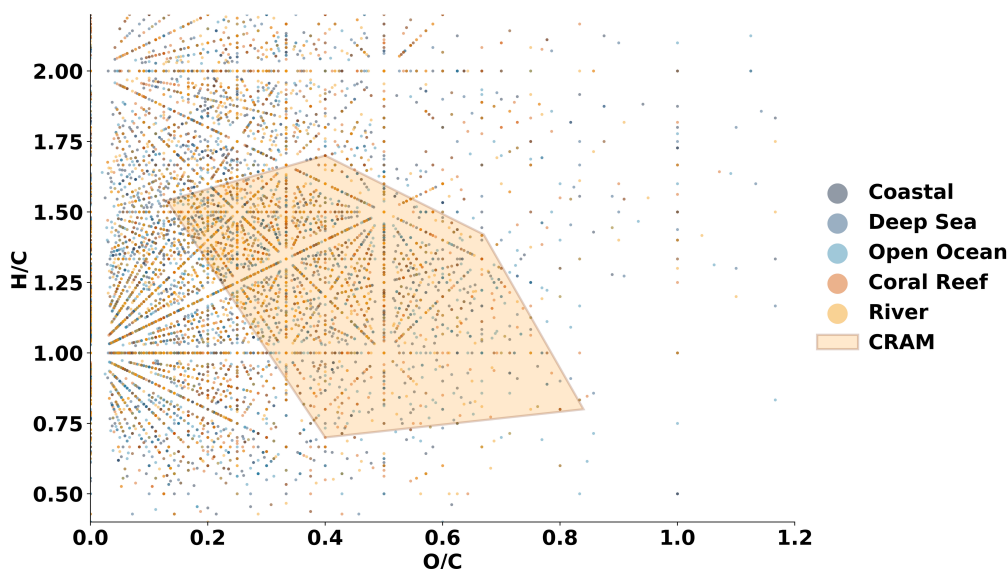
Supplementary Figure S9a. World map of the 98 MassIVE LC–Orbitrap sources.



Supplementary Figure S9b. Broadband DOM spectrum with CRAM-region features highlighted (extended display supporting main Figure 2a).



Supplementary Figure S9c. MS carboxyl-proxy enrichment: CRAM vs non-CRAM nodes by environment.



Supplementary Figure S9d. CRAM-pentagon Van Krevelen stratified by environment.

Formula assignment and extraction of CRAM-region features

This study performed no new collections. All environmental LC–MS/MS spectra are public MassIVE LC–Orbitrap MS¹/MS² deposits. Molecular formulas for the annotation campaign were assigned to high-resolution LC–Orbitrap MS¹ peaks under conservative elemental constraints (C, H, O, N, S, and P). Formula assignment and MS/MS annotation used SIRIUS 5 with CSI:FingerID 5. MS¹ mass-error was tested at 1, 3, 5, and 10 ppm; in each test the MS² fragment-matching tolerance was set to the same ppm as that MS¹ setting. Headline Abstract/Results EPR and coverage statistics are the campaign annotation at 5 ppm (MS¹ = MS² = 5 ppm): TRFU n = 18,513 PubChem-only 31.8%, max-over-union 38.0%, CRAM-only 40.4%. EPR at 1, 3, and 10 ppm is not tabulated. Adducts were hydrogen-based and polarity-matched to each MassIVE dataset: [M+H]⁺ for positive-mode sources and [M-H][−] for negative-mode sources. Elemental constraints remained C, H, O, N, S, and P. Isotope-pattern scoring used the SIRIUS 5 default (no custom mDa window is separately documented). A 5 ppm tolerance in reaction-network construction is a mass-difference edge cutoff, distinct from the SIRIUS MS¹/MS² ppm grid. The CRAM-region restriction should be interpreted as an operational filter for annotation benchmarking rather than a definitive structural classification. At this stage, formulas were selected on the basis of compositional location, after which alternative structural libraries were queried to evaluate how different candidate spaces altered the apparent isomer richness and structural plausibility of the same underlying formulas. This distinction is important because the annotation workflow was designed to compare reference spaces, not to claim direct molecular identification from formula assignment alone.

Candidate retrieval from alternative reference-library scenarios

For each CRAM-region molecular formula, candidate structures were retrieved under multiple reference-library scenarios. The primary comparison was between PubChem alone and PubChem supplemented with the generated CRAM library. Additional control scenarios included COCONUT and ECMDB, which were used to evaluate whether the observed gains were specific

to the CRAM-focused generated library or could be reproduced by supplementing PubChem with other general-purpose natural-product or metabolite resources. For each scenario, the number of candidate isomers per formula was recorded, and the distribution of structural motifs among candidate sets, including ring-system patterns and COOH positioning, was tracked as part of the downstream comparison.

This comparison framework was central to the study design. The objective was not simply to annotate spectra with a larger database, but to test whether adding chemically targeted CRAM-like structures changes the type and quality of candidates returned for formulas occupying a known blind spot in conventional reference collections. In this sense, the annotation workflow was explicitly hypothesis-driven: if CRAM-like structural space is genuinely underrepresented in standard databases, then supplementing PubChem with environmentally plausible CRAM candidates should increase candidate coverage and improve spectral compatibility specifically in the CRAM compositional region.

MS/MS matching and annotation scoring

Environmental MS/MS spectra were matched to candidate structures using fragmentation-tree-based scoring, implemented through SIRIUS/CSI:FingerID (EPR). MetFrag is a separate engine used only in Supplementary Note 10 and does not validate SIRIUS annotations. Formula assignment and MS/MS annotation used SIRIUS 5 with CSI:FingerID 5. MS¹ mass-error was tested at 1, 3, 5, and 10 ppm; in each test the MS² fragment-matching tolerance was set to the same ppm as that MS¹ setting. Headline Abstract/Results EPR and coverage statistics are the campaign annotation at 5 ppm (MS¹ = MS² = 5 ppm): TRFU n = 18,513 PubChem-only 31.8%, max-over-union 38.0%, CRAM-only 40.4%. EPR at 1, 3, and 10 ppm is not tabulated. Adducts were hydrogen-based and polarity-matched to each MassIVE dataset: [M+H]⁺ for positive-mode sources and [M-H]⁻ for negative-mode sources. Elemental constraints remained C, H, O, N, S, and P. Isotope-pattern scoring used the SIRIUS 5 default (no custom mDa window is separately documented). A 5 ppm tolerance in reaction-network construction is a mass-difference edge cutoff, distinct from the SIRIUS MS¹/MS² ppm grid. The synthetic-standard MetFrag used [M-H]⁻ precursors. For each annotation scenario, the workflow recorded the number of candidate isomers per formula, the rank and score of the best-matching structure, and global annotation-performance metrics including explained peak ratio. These outputs were then summarized across environments and database scenarios to determine how expanded CRAM-focused reference space altered the apparent interpretability of CRAM-region spectra.

Effect of CRAM-specific library expansion on annotation depth

The increase in candidate-per-formula values is particularly relevant for the downstream network analysis. Reaction-network reconstruction depends on the availability of structurally plausible nodes in the CRAM region, and sparse reference coverage would otherwise lead to thin or chemically distorted networks. The annotation workflow therefore serves a dual function: it benchmarks the analytical value of the expanded library at the spectral level and also establishes the node density required for subsequent pathway reconstruction. This is why candidate-per-formula enrichment was treated as a key metric alongside explained peak ratio in the present study.

Environmental variation in annotation performance

This distinction is critical for interpretation. The annotation workflow does not establish that a given environment contains more CRAM in an absolute sense. Rather, it shows that the spectra from marine environments are better explained by the expanded CRAM-focused candidate space than those from riverine environments. In the context of this study, that pattern is analytically meaningful because it demonstrates that the generated library is environmentally discriminating rather than uniformly permissive. This property makes it suitable for the subsequent reconstruction of environmentally divergent CRAM formation pathways.

Supplementary Note 5. Reaction-network reconstruction and robustness analyses

Overview

To infer transformation chemistry from the expanded CRAM candidate space, we reconstructed directed reaction networks for each environmental sample using exact mass differences between annotation-supported structural candidates. Because mass-difference networks in ultracomplex mixtures can be inflated by stoichiometrically valid but chemically implausible edges, the network analysis in this study was designed around two complementary principles: first, candidate transformations had to satisfy a predefined reaction-mass rule set; second, each proposed transformation had to pass an independent plausibility check based on atom-mapping consistency. The resulting workflow was intended to retain network interpretability while minimizing chemically unrealistic connectivity.

In the main manuscript, this network layer provides the reaction-level component of the integrated analytical framework and supports the conclusion that CRAM formation is oxidation-rich but environmentally divergent. The purpose of the present Supplementary Note is therefore to document how the networks were constructed, why the RXNMapper filtering step was used, how network topology varied across environments, and what analytical caveats remain in interpreting network-derived transformations.

Reaction-space definition and candidate edge construction

Reaction networks were constructed separately for each of the 98 environmental sources. Candidate edges were assigned between any two annotation-supported structures whose exact mass difference matched one of 27 predefined biochemical reaction classes within a tolerance of 5 ppm. These reaction classes included oxidation, reduction, decarboxylation, dehydration, methylation, hydroxylation, carboxylation, alkylation, acylation, condensation, glycosylation, and additional related mass-difference motifs used throughout the study. Environment-resolved formation-edge class frequencies are summarized in Supplementary Table S2; RXNMapper edge-level scores are in Supplementary Table S11.

Directionality was assigned according to the sign of the mass change relative to the reference reaction mass, allowing each edge to be interpreted as a forward or reverse transformation. Under this scheme, a given edge could contribute to CRAM formation, CRAM degradation, or internal cycling depending on the classification of the source and target nodes under the annotation framework. This directional formulation allowed the same reaction catalog to be used consistently across all environments and later enabled decomposition of the networks into formation, degradation, and internal reaction modes.

RXNMapper-based plausibility filtering

Exact mass matching alone is insufficient to identify chemically plausible reactions, particularly in large annotation-derived networks where numerous candidate edges can satisfy the same stoichiometric constraint. To reduce this ambiguity, each candidate transformation was evaluated using RXNMapper, a transformer-based atom-mapping model trained on large corpora of organic reactions. RXNMapper assigns a confidence score between 0 and 1, reflecting the internal consistency of the proposed bond-breaking and bond-forming pattern with learned reaction chemistry. In the present workflow, edges with RXNMapper confidence below 0.5 were discarded, and only edges supported by both stoichiometric mass balance and mapping plausibility were retained for downstream analysis.

Directed mass-difference networks were built for each of the 98 sources. Candidate edges match one of 27 predefined reaction classes within a 5 ppm mass-difference tolerance (network-edge tolerance only, not an MS1 formula cutoff) and are retained at RXNMapper confidence ≥ 0.5 , yielding 878,941 directional edges (1,550,650 at the score-distribution threshold).

A more stringent subset of high-confidence edges, defined here as RXNMapper > 0.8 , comprised 23.4% of the retained network and showed 2.1-fold enrichment for oxidation/reduction reactions relative to the full retained set. This enrichment is notable because it indicates that the oxidative bias observed in the main results is not driven primarily by weakly supported reaction assignments. Rather, the most confidently mapped transformations remain disproportionately associated with oxidation-related chemistry, supporting the interpretation that oxidative processing is a robust feature of CRAM-associated transformation space.

Node classification and decomposition of network modes

After plausibility filtering, network nodes were categorized as CRAM-like or non-CRAM-like according to the best-scoring structural annotation associated with each node. This classification enabled decomposition of each sample-specific network into three directional modes: non-CRAM- $\rightarrow$ CRAM formation edges, CRAM- $\rightarrow$ non-CRAM degradation edges, and internal reactions between nodes of the same class. Under this annotation-defined framework, the main manuscript reported 278,177 formation edges, 180,419 degradation edges, and 735,961 internal non-CRAM edges across the integrated analysis.

This classification step is analytically useful but also represents an important caveat. Because CRAM/non-CRAM status depends on the best-scoring annotation rather than direct structural confirmation, the resulting network partitions should be interpreted as annotation-supported network modes rather than experimentally confirmed reaction classes. In particular, enrichment analyses comparing CRAM and non-CRAM subnetworks are conditional on the current annotation framework and therefore reflect the behavior of the expanded candidate space under the present scoring logic. Throughout the manuscript, we interpret these subnetworks as evidence-bearing analytical constructs rather than definitive molecular identities.

Network topology across environments

After filtering, network topology differed systematically across environments. River-influenced samples exhibited the highest network density (0.0415) and the highest average node degree (12.44), consistent with strongly connected transformation series typical of terrestrially derived

homologous mixtures. Coral reef environments showed the lowest density (0.0109) and lowest average degree (3.25), consistent with a more heterogeneous and less recurrent set of candidate transformation relationships. Coastal, deep-sea, and open-ocean samples occupied an intermediate but internally coherent regime, with high pairwise inter-environment similarity, including 0.989 between deep sea and open ocean and 0.979 between coastal and deep sea. By contrast, river networks showed consistently lower similarity to all marine systems, ranging from 0.487 to 0.608.

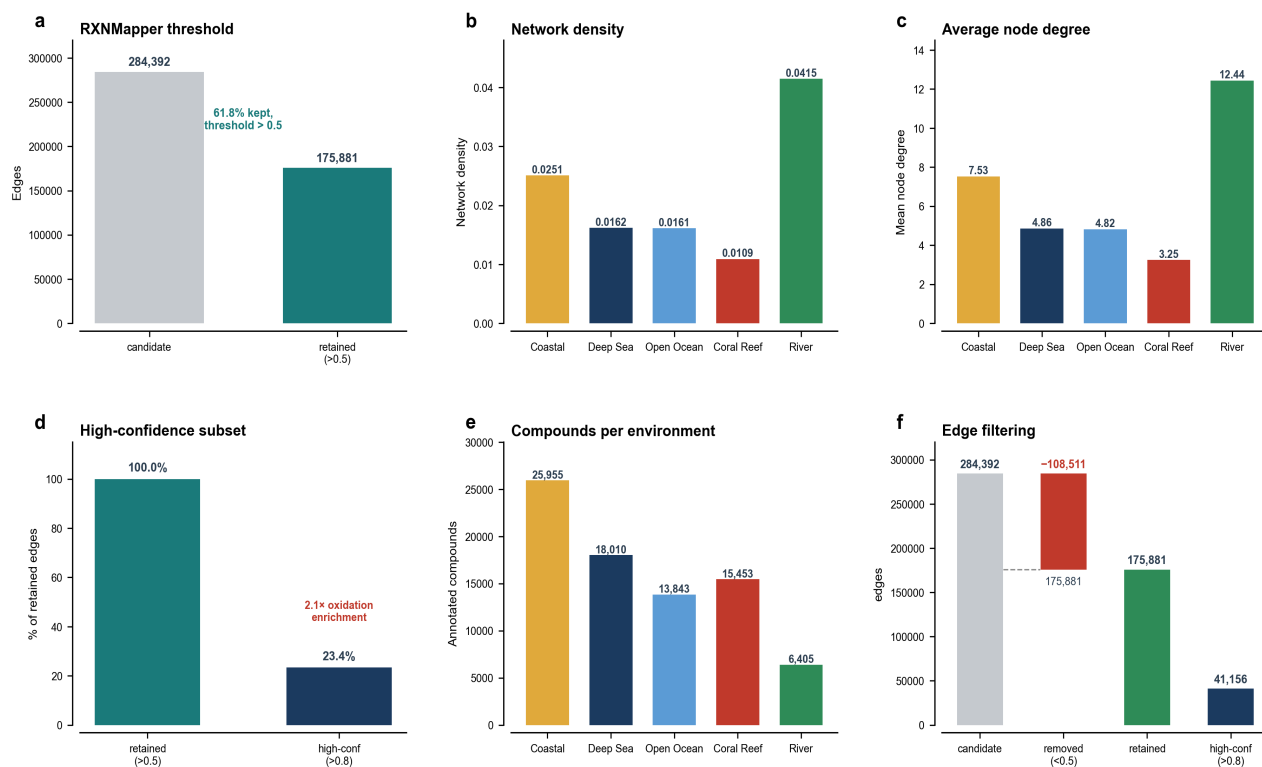
These topological differences support two important interpretations. First, the marine pelagic environments form a relatively coherent transformation domain despite geographic and depth differences, consistent with shared oxidative processing regimes across the marine DOM continuum. Second, riverine networks are not merely weaker versions of marine networks but occupy a topologically distinct space characterized by denser, more connected transformation series. This distinction is important because it suggests that the environmental divergence described in the main text emerges not only from pathway composition but also from the overall structure of network connectivity.

Reaction-category robustness and confidence structure

RXNMapper confidence and directional-similarity scores varied systematically by reaction class and environment. Among major categories, acylation and glycosylation showed the highest mean plausibility scores, whereas oxidation, despite being the most abundant category, showed somewhat lower mean confidence, reflecting the greater structural heterogeneity of oxidative transformations in the candidate set. At the environment level, river networks showed the highest overall mapping confidence and directional similarity, whereas deep-sea networks showed the lowest. This pattern is consistent with a simpler and more chemically familiar transformation space in riverine samples and a more structurally novel, annotation-challenging space in deep-ocean DOM. Importantly, lower mean confidence for oxidation should not be interpreted as evidence against oxidation-driven chemistry. Rather, it likely reflects the broader chemical diversity of oxidation-like transformations, which can operate on many distinct scaffolds and functional-group contexts. The fact that oxidation remains dominant even after plausibility filtering, and is additionally enriched in the high-confidence subset, argues that its prominence is not an artifact of a permissive reaction catalog. Instead, oxidation appears to be both common and chemically heterogeneous, which is consistent with its central role in environmental organic-matter processing.

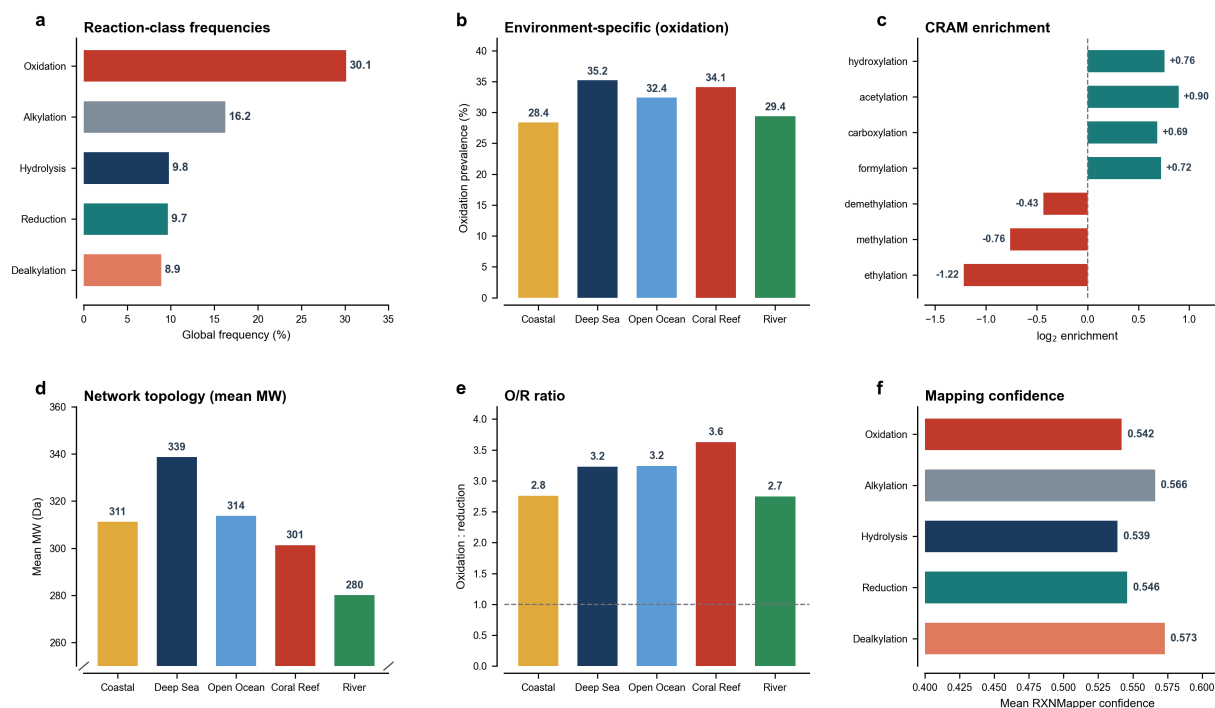
Relationship to the main reaction-level conclusions

The robustness analyses presented here support the key reaction-level conclusions of the manuscript in three ways. First, the network is substantially reduced by plausibility filtering, demonstrating that the final retained connectivity is not a trivial consequence of mass-difference matching alone. Second, the environmental topology remains structured after filtering, with coherent pelagic marine similarity and distinct riverine connectivity. Third, oxidation-related reactions remain central under both the retained and high-confidence subsets, supporting the interpretation that oxidation-rich CRAM formation is a robust property of the annotation-supported network rather than a threshold artifact. Reaction-level divergence is meaningful only if the network itself is chemically constrained. By integrating exact mass-difference rules with RXNMapper-based plausibility scoring, the present workflow provides a conservative but interpretable network substrate on which those environmental comparisons can be made.



Supplementary Figure S10 — Robustness of Reaction-Network Reconstruction

(a) confidence threshold effect; (b) network density; (c) node degree; (d) retained edges; (e) environmental comparison; (f) sensitivity summary.



Supplementary Figure S11 — Additional Reaction-Network Analyses

(a) reaction-class frequencies; (b) environment-specific networks; (c) reaction enrichment; (d) network topology; (e) oxidation/reduction comparison; (f) mapping-confidence by class.

Supplementary Note 6. Sequential reaction ordering and Markov analyses

Overview

The reaction-network analyses described in Supplementary Note 5 establish which reaction categories are prevalent in CRAM-associated transformation space. However, reaction prevalence alone does not indicate whether transformations occur in ordered sequences or merely co-occur within the same network. To address this, we constructed first-order Markov models from the directed reaction networks and used them to quantify sequential reaction preferences across environments and network modes. The aim of this analysis was to determine whether CRAM formation and transformation are associated with recurrent reaction orderings, and whether those orderings differ systematically between marine and riverine systems.

In the main manuscript, this analysis supports the interpretation that CRAM formation is not only oxidation-rich, but also pathway-structured. The present Supplementary Note documents how the transition matrices were constructed, how plausibility weighting was incorporated, what the dominant sequential patterns were, and what interpretive boundaries should be maintained when reading reaction-order results from annotation-supported networks.

Construction of first-order Markov chains

First-order Markov chains were constructed separately for each sample-specific directed reaction network. For every intermediate node, defined here as a node with at least one incoming edge and

at least one outgoing edge, we recorded the ordered pair of reaction categories corresponding to the incoming transformation A and the outgoing transformation B. Each such A to B pair was treated as one empirical observation of sequential reaction ordering. In this way, the network was converted into a set of local reaction transitions that capture how one transformation category tends to be followed by another at intermediate molecular states.

Transition probabilities $P(B | A)$ were estimated by normalizing weighted path counts: for each pair of states A and B, the weighted count was summed over all intermediate nodes i lying on a path from A to B, using weights $w_i(A, B) = c_{in,i} \times c_{out,i}$, the product of the RXNMapper confidence scores for the incoming and outgoing reaction-mapping edges at node i . This sum was then divided by the total weighted count summed over all possible destination states B' , yielding a properly normalized transition probability distribution. This weighting scheme was introduced to give greater influence to sequential transitions supported by two chemically plausible steps rather than by a single high-confidence edge paired with a weakly supported one. The resulting transition matrix T , with entries $T_{AB} = P(B | A)$, summarizes the preferred reaction ordering within a given network or subnetwork.

Transition matrices were computed separately for the major directional modes of the reaction framework, including CRAM formation, CRAM degradation, and internal cycling subnetworks, and were then compared across environments. This decomposition was important because the same reaction categories can play different mechanistic roles depending on whether they occur during entry into CRAM space, exit from CRAM space, or internal CRAM-to-CRAM transformation.

Dominant sequential patterns in marine formation networks

Across marine formation subnetworks, oxidation-to-carboxylation is a minority first-order Markov branch ($P/(1/27)=0.513$), not a strong tendency. Marine formation $P(\text{carboxylation} | \text{oxidation})$ 0.019 (range 0.014–0.023; Table 3), below a uniform $1/27$ 0.037 baseline among 27 classes. The oxidation-to-carboxylation branch is a minority topological ordering statistic, not a kinetic signature and not enzymatic shielding. This result is consistent with the reaction-frequency analysis in the main text, in which oxidation-related mass differences dominated CRAM formation and carboxylation directly contributed the carboxyl groups characteristic of CRAM-like structures. In this sense, the Markov analysis adds an ordering dimension to the same mechanistic picture: oxidation and carboxylation do not merely co-occur, but tend to occur in sequence.

This oxidation-to-carboxylation preference is also consistent with the dominant formation motifs identified in the directional network analysis, including hydroxylation–carboxylation and formylation–carboxylation sequences. The Markov formulation therefore provides a compact quantitative description of a pathway logic that is already hinted at in the reaction catalog: marine CRAM formation tends to proceed through oxidative functionalization rather than through reductive simplification or alkyl-chain growth.

Fragmentation-associated transitions and oxidative clustering

The sequential analysis also identified characteristic patterns associated with CRAM breakdown and scaffold rearrangement. Decarboxylation $\rightarrow$ ring opening is $P = 0.00$ across pathway types. This pattern is consistent with the broader network observation that CRAM degradation in marine

systems is itself oxidation-dominated and often associated with oxidative fragmentation rather than reductive deoxygenation.

Self-transition probabilities further clarified the structure of marine reaction sequences. Among the major categories, oxidation showed the highest self-transition probability, $P(\text{ox}|\text{ox})$ 0.38 ± 0.05 , indicating that oxidative steps tend to cluster rather than appearing as isolated events. This pattern should not be interpreted literally as repeated identical biochemical reactions on a single known substrate. Rather, within the present annotation-supported network framework, it indicates that molecular intermediates generated by one oxidation-related transformation are particularly likely to participate in a subsequent oxidation-related transformation. The result is consistent with progressive oxidative remodeling as a recurring feature of marine CRAM transformation space.

Divergent sequential logic in riverine networks

Riverine networks showed a markedly different transition structure. Whereas marine formation $P(\text{carboxylation} | \text{oxidation})$ 0.019 remains below $1/27$, riverine alkylation-to-hydrolysis in CRAM formation remains $P = 0.099 \pm 0.177$ as in the main text. This pattern is consistent with the distinct reaction frequencies and denser topology observed for riverine networks in Supplementary Note 5, and supports the interpretation that riverine CRAM-like transformation space is more strongly influenced by fragmentation and modification of heterogeneous biopolymer-derived precursors. In other words, the contrast between marine and riverine systems is not limited to differences in reaction abundance; it is also expressed in the preferred ordering of transformation categories.

This sequential divergence is especially important for the manuscript's central framing. If aquatic environments differed only in the relative abundance of reaction types, then CRAM pathway divergence might be interpreted as a weak compositional shift within a largely shared transformation logic. The Markov results argue against that interpretation. Instead, they indicate that the reaction pathways themselves are ordered differently across environments, reinforcing the conclusion that CRAM formation follows environmentally divergent trajectories.

Relationship to the integrated pathway interpretation

The sequential ordering results strengthen the main analytical framework in two ways. First, they show that reaction-network structure contains interpretable pathway information beyond simple edge counts. Second, they provide a mechanistic bridge between reaction-level observations and the biosynthetic provenance analyses presented later in the manuscript. For example, the marine preference for oxidation-to-carboxylation transitions is consistent with the enrichment of terpenoid-associated precursors, whereas the riverine preference for alkylation-to-hydrolysis is consistent with amino acid- and polysaccharide-influenced precursor pools. Thus, the Markov analysis helps connect reaction ordering to broader differences in precursor landscape.

At the same time, the Markov formulation remains intentionally conservative. It does not claim to reconstruct complete biochemical pathways or to establish direct temporal sequences in the experimental system. Rather, it summarizes the most probable local ordering of reaction categories within the annotation-supported network. Within that evidentiary tier, the results are informative because they show that the network does not behave like an unordered mass-difference graph; instead, it contains structured directional motifs that differ across environmental settings.

Supplementary Note 7. NPClassifier resemblance labels and pathway tables

Overview

To complement the reaction-network analysis, we assigned biosynthetic context to annotation-supported candidate structures using NPClassifier and examined how pathway composition changed across precursor, intermediate, and end-product stages of CRAM-associated networks. The purpose of this analysis was not to infer exact biosynthetic histories for individual environmental molecules, but to determine whether CRAM formation is preferentially associated with particular regions of biosynthetic structural space, and whether those associations differ systematically across aquatic environments. In the main manuscript, this layer supports the interpretation that environmentally divergent CRAM formation pathways are linked not only to reaction chemistry, but also to precursor identity and downstream persistence.

Within the integrated framework of the study, biosynthetic provenance analysis serves as the interpretive bridge between structural candidates and environmental pathway logic. The reaction-network layer identifies which classes of transformations are prevalent and how they are sequentially organized, whereas the provenance layer asks from which kinds of molecular precursor pools those transformations are most likely to originate. Because the present workflow is based on annotation-supported candidate structures, all provenance assignments should be interpreted as classifier-supported structural context rather than experimentally confirmed biosynthetic origin. Even with this caveat, the pathway distributions provide a useful and internally consistent way to compare CRAM-associated molecular space across environments.

NPClassifier workflow and scope of interpretation

All unique candidate structures observed across the 98 environmental sources were classified using NPClassifier, which assigns molecules to a hierarchical organization of pathway, superclass, and class categories on the basis of structural fingerprints. For the main analyses, the pathway-level assignments were used as the primary biosynthetic descriptor because they provide the broadest and most robust summary of precursor landscape. Superclass- and class-level tables can nevertheless be retained in the Supplementary Information for completeness and for readers interested in finer taxonomic resolution.

For each network node, the pathway label of the best-scoring structural candidate was used. Pathway distributions were then summarized separately for three analytically defined stages: precursor nodes, intermediate CRAM nodes, and end-product CRAM nodes. In addition, pathway shifts were examined within non-CRAM to CRAM formation edges by comparing the pathway composition of source versus target nodes. This allowed us to assess whether specific biosynthetic categories were preferentially enriched or depleted during entry into CRAM space. Because node labels depend on the annotation framework and because NPClassifier assigns structural similarity rather than true biosynthetic lineage, the resulting pathway distributions are best interpreted as patterns of biosynthetic resemblance within the candidate space.

CRAM nodes are alkaloid-, terpenoid-, and shikimate-structured (Supplementary Table S3)

Across environments, annotation-supported CRAM nodes match Supplementary Table S3 (means $\pm$ SEM). Alkaloid-associated labels are a major CRAM-node class and often the largest marine share (coastal $24.04 \pm 1.41\%$; deep sea $23.61 \pm 1.80\%$; open ocean $27.72 \pm 2.02\%$; coral reef 27.18

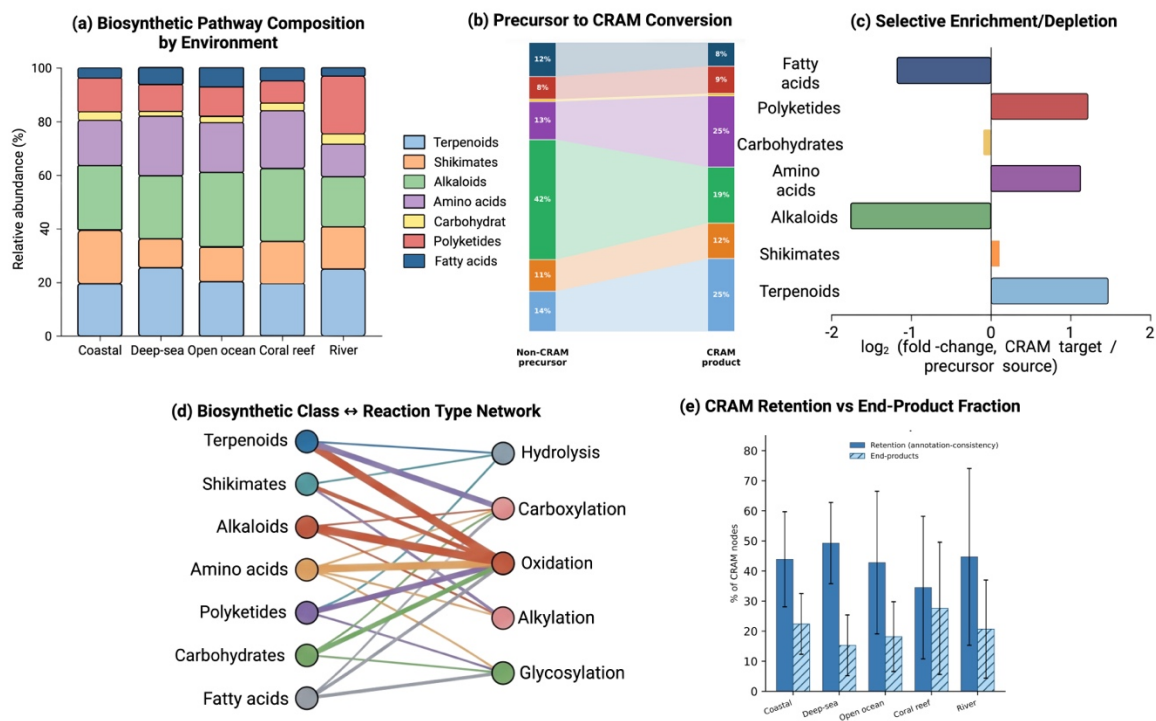
$\pm 2.84\%$; river $18.57 \pm 3.43\%$). Terpenoid-associated labels are comparable ($19.65 \pm 1.27\%$ coastal; $25.65 \pm 3.34\%$ deep sea; $20.48 \pm 1.95\%$ open ocean; $19.61 \pm 2.43\%$ coral reef; $25.17 \pm 5.07\%$ river) but are not a sole majority. Shikimate-associated labels contribute $19.97 \pm 2.00\%$ (coastal), $10.55 \pm 1.22\%$ (deep sea), $12.86 \pm 2.14\%$ (open ocean), $15.79 \pm 2.80\%$ (coral reef), and $15.75 \pm 2.11\%$ (river). These percentages are NPClassifier labels on annotation-supported nodes, not evidence of a single biosynthetic precursor pool. Riverine CRAM differed most clearly in elevated polyketide-associated labels ($21.73 \pm 1.45\%$ versus $8.34\text{--}12.54\%$ across marine environments in Panel C), with amino acids $12.18 \pm 2.29\%$ and carbohydrates $3.71 \pm 1.37\%$. Carbohydrate-associated labels remain a minority class in all environments in Panel C (marine $1.69\text{--}3.18\%$; river 3.71%). Together, these distributions indicate an alkaloid-inclusive, multi-pathway CRAM-node mix rather than a terpenoid-only story.

Source-to-target pathway shifts during CRAM formation

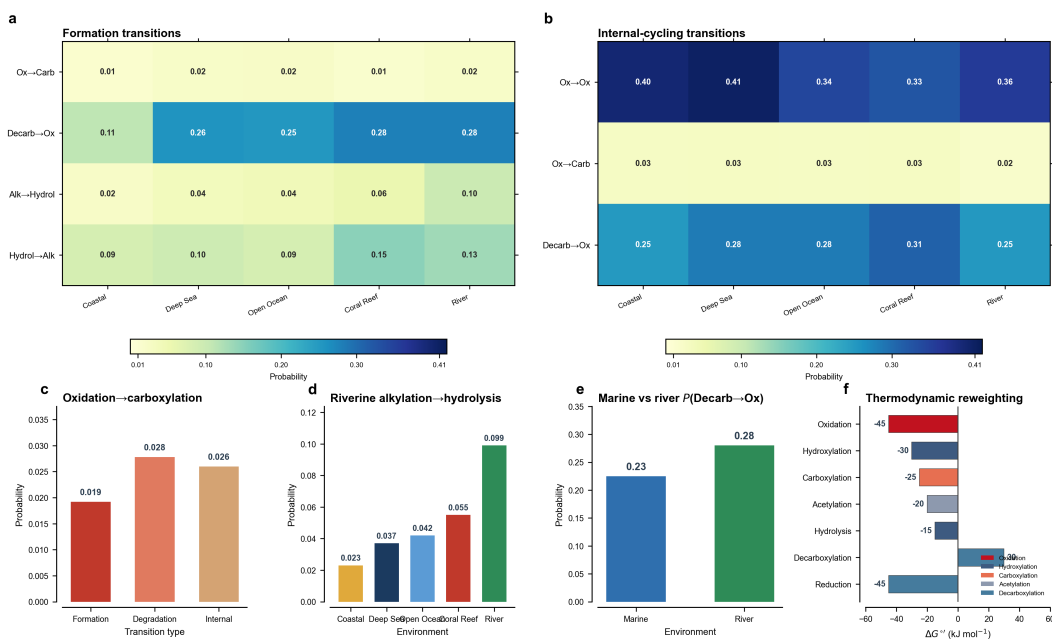
To determine how biosynthetic composition changes during entry into CRAM space, we examined pathway distributions along non-CRAM to CRAM formation edges. Across the pooled 98-network formation-edge set ($n = 118,336$ edges), alkaloid-associated features represented 41.6% of formation sources but only 19.4% of formation targets, whereas terpenoid-associated structures rose from 13.9% to 25.2% (1.82-fold) and amino acid-associated labels rose from 13.2% to 24.7% . Alkaloid-like resemblance is therefore emphasized on the precursor side of CRAM formation as well as remaining a major Panel C CRAM-node class, while terpenoid-like resemblance is selectively enriched among CRAM targets. These pooled shifts are NPClassifier resemblance statistics along mass-difference edges, not proof of specific biochemical deamination or denitrogenating reactions.

Relationship between provenance and environmentally divergent formation pathways

Taken together, the provenance analyses strengthen the central claim of the manuscript in two ways. First, they show that CRAM-associated candidate space is alkaloid-inclusive at the node level (Supplementary Table S3) and that pooled formation edges deplete alkaloid-like labels ($41.6\% \rightarrow 19.4\%$) while enriching terpenoid-like labels ($13.9\% \rightarrow 25.2\%$), not only that reaction frequencies differ. Riverine Supplementary Table S3 space is distinguished especially by elevated polyketide-associated labels. Second, these provenance differences sit alongside annotation-consistency and end-product indices (Supplementary Table S8) that remain source-heterogeneous rather than a simple marine-versus-river residence-time regime.



Supplementary Figure S11a. Persistence, end-product fractions, and sequential remodeling (annotation-consistency is Supplementary Table S8).



Supplementary Figure S11b. Markov transition probabilities and thermodynamic reweighting across environments.

(a)–(b) formation and internal-cycling transition heatmaps; (c) oxidation→carboxylation by mode; (d) riverine alkylation→hydrolysis; (e) marine vs river decarboxylation→oxidation; (f) Delta G' reweighting of reaction classes.

Supplementary Note 8. Additional cross-environment analyses and caveats

Overview

The main manuscript reports environmental divergence in CRAM formation supported by deposited tables: oxidation-rich reaction chemistry across environments (Table 3 Panel B), riverine elevation of alkylation and alkylation-to-hydrolysis Markov contribution (Tables 2–3), and environment-specific NPClassifier pathway mixes (Supplementary Table S2-3). Because these interpretations rest on a multistep annotation-supported workflow and because sample numbers are uneven across environments, it is important to evaluate how strongly the cross-environment contrast is supported and where caution should remain explicit. The purpose of this Supplementary Note is therefore to consolidate the additional cross-environment analyses, to clarify the evidentiary status of the freshwater–marine contrast, and to define the principal interpretive boundaries of the study.

Coral reef as an intermediate or transitional CRAM regime

Additional cross-environment analysis indicates that coral reef samples occupy a distinct position relative to the pelagic marine continuum. In the main network summaries, coral reef CRAM showed reduced oxidation dominance and a somewhat different precursor composition. Composite z-score analysis across six network metrics identified coral reef as the most anomalous environment, with a composite $|z|$ score of 2.4. Reaction dynamics were similarly distinct: oxidation dominance in coral reef samples was lower than in other marine systems. Coral-reef $n = 4$ is a sampling limitation.

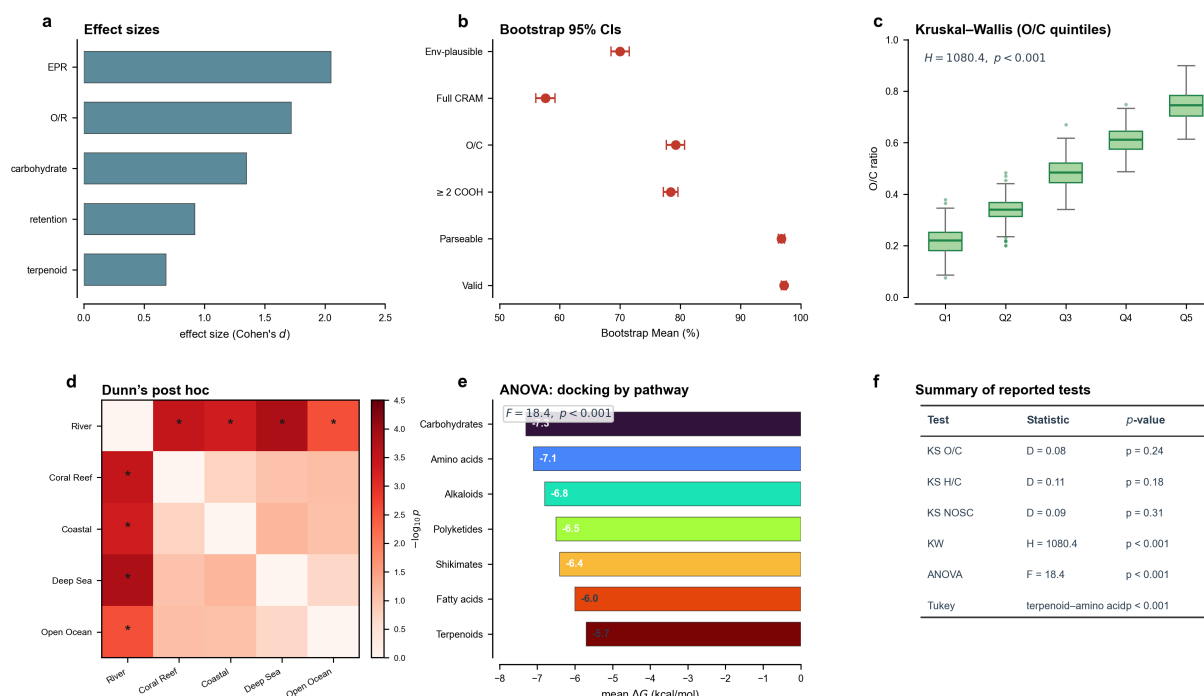
SPE recovery bias and robustness of cross-environment ordering

A further concern in cross-environment comparison is the possibility that apparent environmental gradients are shaped in part by analytical artifacts, especially differences in solid-phase extraction recovery across DOM compound classes. To address this, a systematic SPE bias correction analysis was performed using published recovery factors for major DOM classes. In this analysis, CRAM-like molecules were assigned a recovery estimate of $RCRAM=0.72$, whereas non-CRAM molecules were assigned a weighted-average recovery of $R_{other}=0.45$. Under these assumptions, absolute CRAM fractions were uniformly reduced after correction. For example, coastal estimates decreased from 0.35 to 0.24, deep-sea estimates from 0.33 to 0.22, and riverine estimates from 0.08 to 0.04.

Crucially, the relative ordering of environments was preserved after correction, and the qualitative marine–freshwater contrast remained significant across the tested recovery scenarios. Sensitivity analysis across a range of recovery assumptions ($RCRAM=0.5–0.9$; $R_{other}=0.3–0.6$) indicated that observed CRAM fractions may overestimate absolute values by approximately 15–45%, but did not overturn the main qualitative conclusion that marine environments are more strongly represented in annotation-supported CRAM space than riverine environments. This result is important because it shows that the environmental contrast reported here is not a trivial consequence of one reasonable but biased extraction scenario. Rather, the contrast is robust in ordering even if absolute class fractions are systematically shifted.

Interpreting retention: network-topological persistence versus environmental longevity

The retention metric used throughout this study should be interpreted carefully. CRAM retention was defined as the fraction of outgoing CRAM edges that connect to other CRAM nodes rather than to non-CRAM nodes. Live annotation-consistency is Supplementary Table S8 only (coastal $43.9 \pm 15.8\%$; deep sea $49.3 \pm 13.5\%$; open ocean $42.8 \pm 23.7\%$; coral reef $34.5 \pm 23.7\%$; river $44.7 \pm 29.4\%$). River intervals overlap coastal and open-ocean intervals. The metric is not residence time. However, retention in this framework does not equal environmental residence time.



Supplementary Figure S12 —Dataset Statistical Validation

(a) effect sizes; (b) confidence intervals; (c) Kruskal-Wallis; (d) Dunn's post hoc; (e) ANOVA; (f) summary of all tests

Supplementary Note 9. Quantum-chemical reactivity profiling of CRAM candidates

Two-tier reactivity screen. Library-wide conceptual-DFT reactivity scores cover 952,210 unique generated CRAM-like candidates. PySCF frontier-orbital DFT (B3LYP-D3/6-31G(d)) was run on a deposited subsample ($n = 79,808$).

PySCF-B3LYP-D3 framework for CRAM reactivity

Electronic reactivity descriptors were computed for generated CRAM candidates using a density-functional workflow in PySCF at the B3LYP-D3/6-31G(d) level with Grimme's empirical dispersion correction (D3). These are in silico descriptors, not measured environmental rates. This level of theory was selected as a practical compromise between chemical interpretability and tractability for very large molecular sets, while preserving established conceptual-DFT (cDFT) descriptors relevant to redox-sensitive organic matter. For each molecule, frontier-orbital energies

were used to compute ionization potential, electron affinity, chemical potential, hardness, softness, electronegativity, and electrophilicity index.

Local susceptibility was summarized by atom-centered electrophilic and nucleophilic site indices together with dual-descriptor extremes, enabling joint analysis of global molecular softness and localized reactive hotspots. This is particularly relevant for CRAM because environmentally persistent alicyclic frameworks can still undergo repeated functional-group-level oxidation if local electronic activation is spatially concentrated at oxygenated substituent environments.

Operationally, this cDFT framework was used as an electronic-response map rather than a direct simulation of seawater reaction kinetics. The descriptor set is intended to rank intrinsic susceptibility across structurally diverse CRAM candidates under a common quantum-chemical reference, so that electronic trends can be compared with network-scale transformation frequencies. In this context, higher softness and lower hardness are interpreted as greater propensity for charge-redistributing oxidative steps, while dual-descriptor extremes indicate where those steps are most likely to localize on a given scaffold.

Global reactivity statistics and structure–reactivity coupling

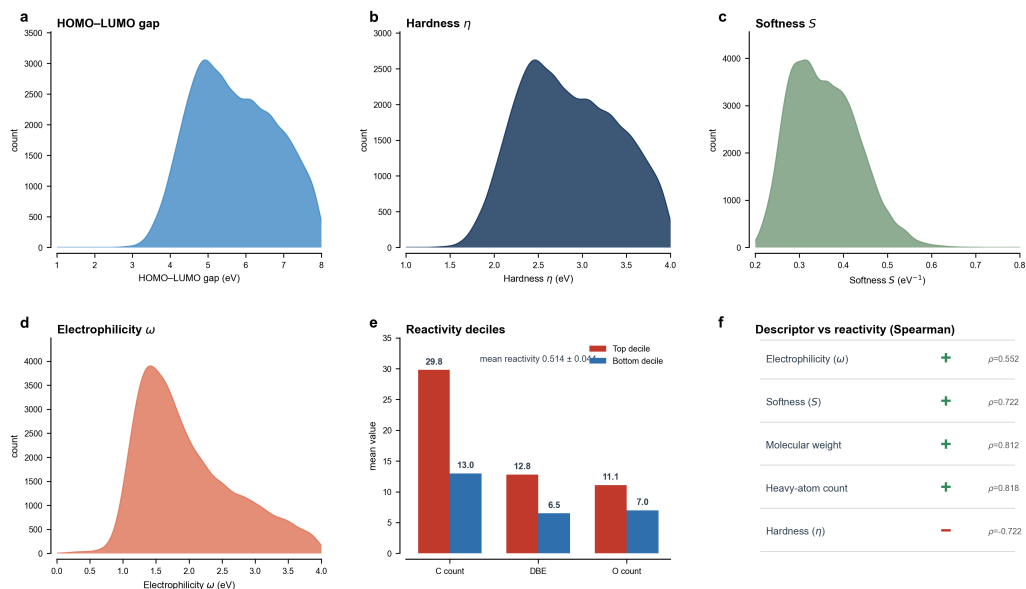
Across 952,210 unique CRAM candidates, mean molecular reactivity probability was 0.514 ± 0.044 (90th percentile: 0.579), while hotspot intensity remained strongly right-skewed (mean 0.246; median 0.020; 90th percentile: 0.873). This distribution indicates that the CRAM candidate space contains many molecules with moderate average reactivity but sharply localized high-reactivity atomic sites, consistent with stepwise edge-functionalization chemistry rather than global scaffold destabilization.

To further interpret this coupling, we considered descriptor directionality across structural complexity classes. The joint increase of mean reactivity with molecular size (molecular weight and heavy-atom count) and with softness indicates that expansion of oxygenated substituent topology does not simply dilute reactive probability across larger graphs; instead, it can create electronically cooperative environments in which multiple nearby heteroatom centers sustain repeated oxidation-compatible transitions. This is consistent with the observed right-skew in hotspot intensity, where most molecules have moderate global reactivity but a subset contains strongly activated local sites that can dominate pathway entry.

Mechanistic implications for CRAM persistence

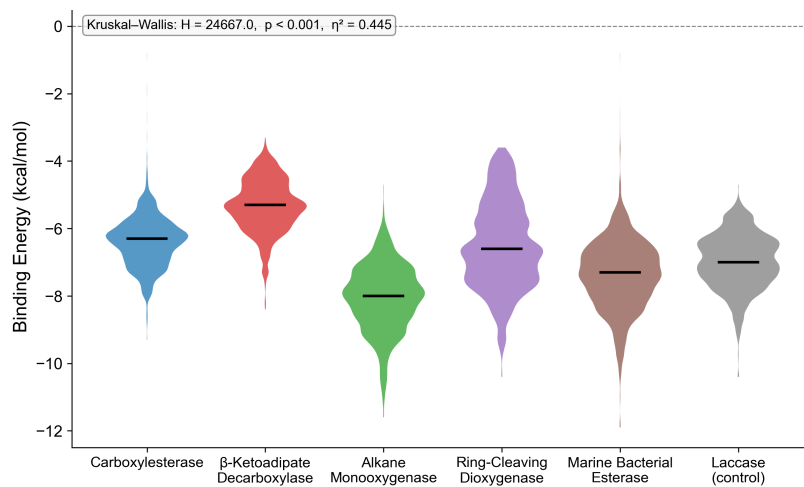
The combined cDFT and hotspot results suggest a dual-scale reactivity regime in CRAM evolution: (i) globally, molecules remain within an electronically feasible oxidation window characterized by moderate mean reactivity; (ii) locally, specific atom-centered sites exhibit high susceptibility, enabling repeated selective transformations. This architecture is compatible with observed CRAM behavior in which molecules continue to transform through oxidation/carboxylation cycles without immediate collapse of alicyclic scaffold integrity.

Accordingly, the reactivity analysis supports the interpretation that CRAM persistence does not require chemical inertness. Instead, persistence can emerge from a balance between localized functional-group turnover and retention of structurally stable, oxygenated backbone motifs. This provides a mechanistic bridge between molecular-electronic reactivity descriptors and the source-resolved network evidence for oxidation-rich yet environmentally persistent CRAM dynamics.

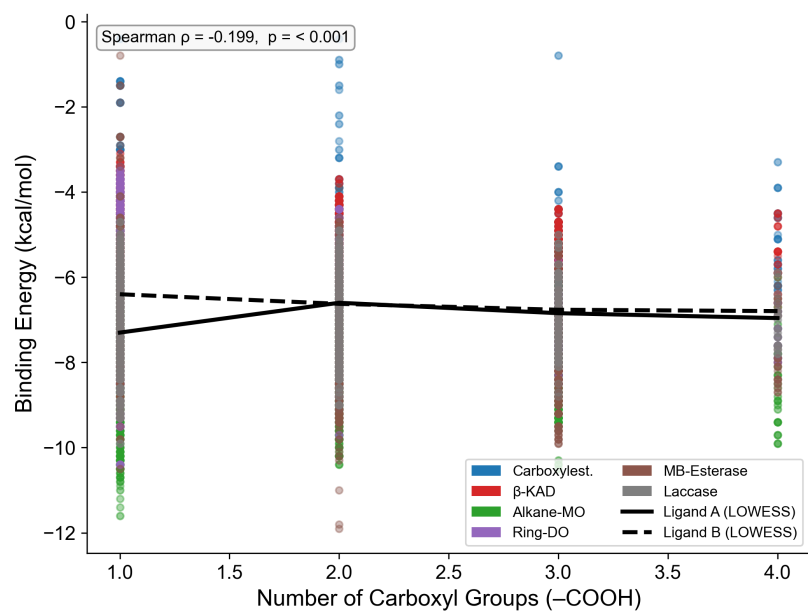


Supplementary Figure S13 — Quantum-Chemical Validation

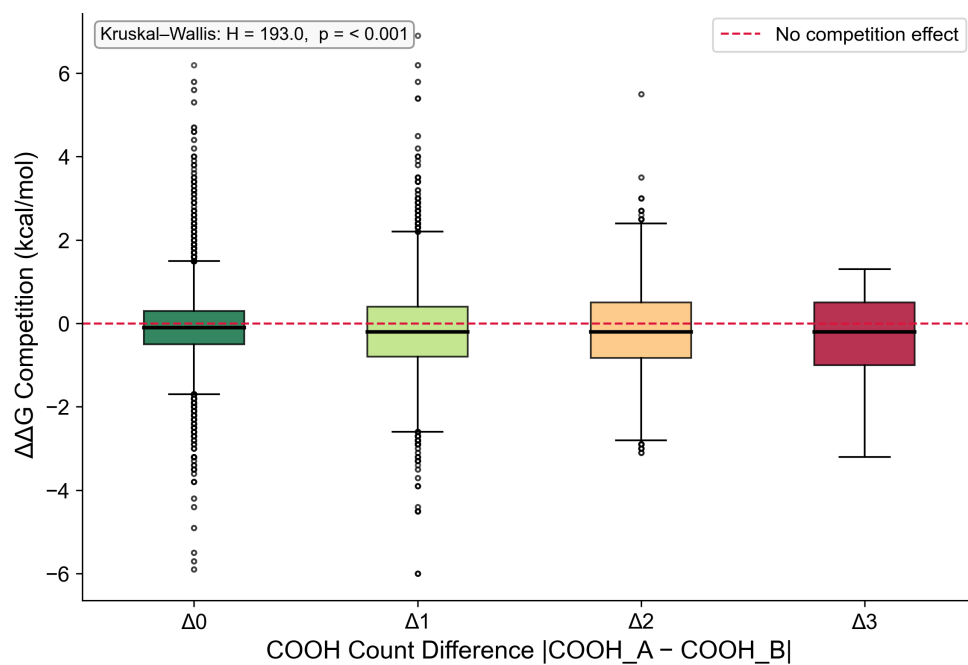
(a) HOMO-LUMO; (b) hardness; (c) softness; (d) electrophilicity; (e) reactivity hotspots/decile comparison; (f) descriptor correlations



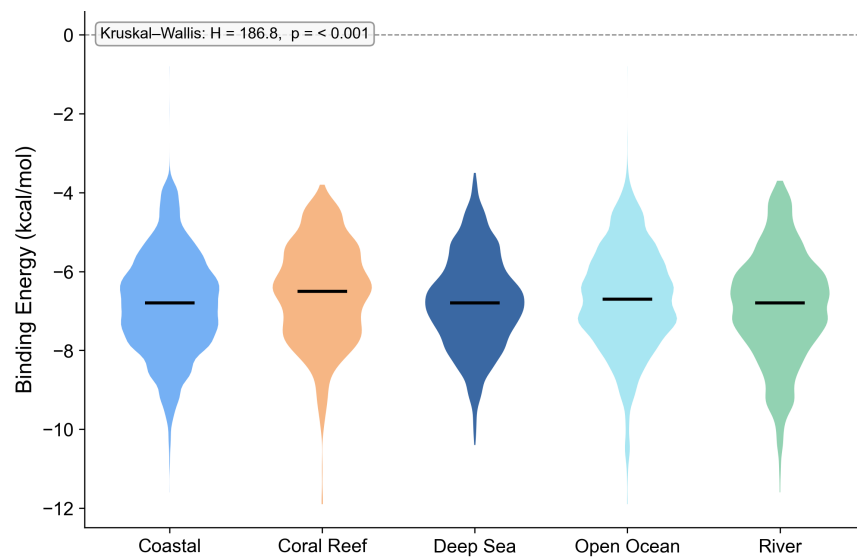
Supplementary Figure S14a. Binding free energy by enzyme (competitive A+B poses).



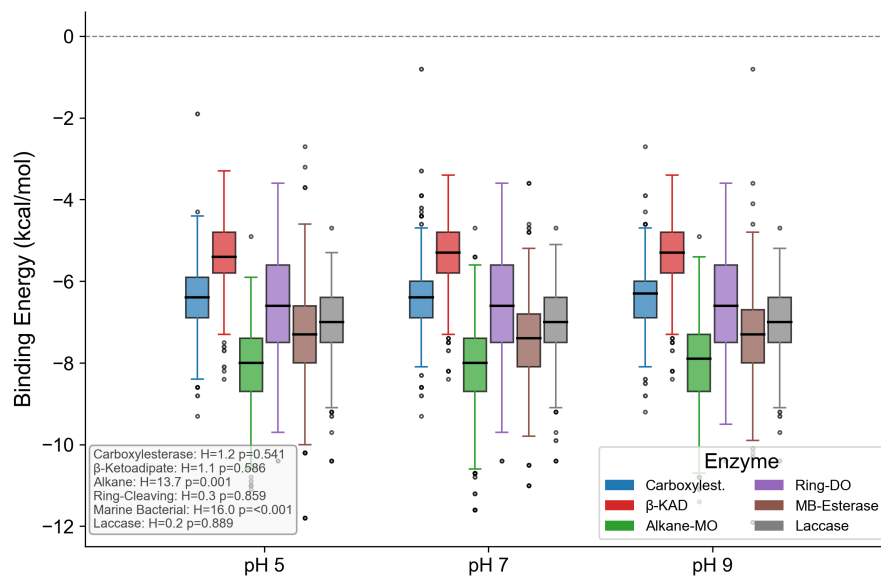
Supplementary Figure S14b. Binding free energy versus carboxyl count (nCOOH).



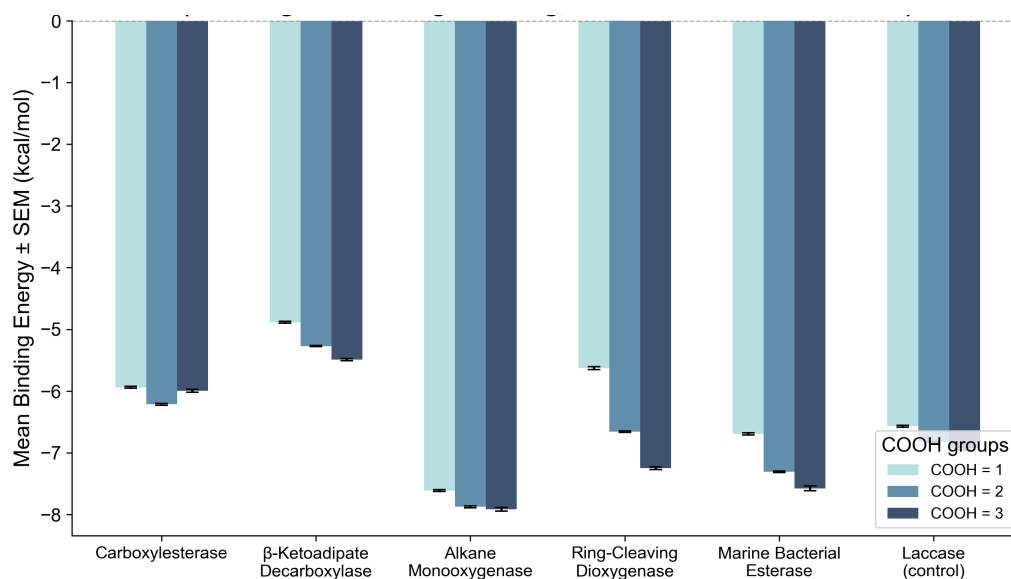
Supplementary Figure S14c. Pair DeltaDeltaG versus carboxyl-count difference along network edges.



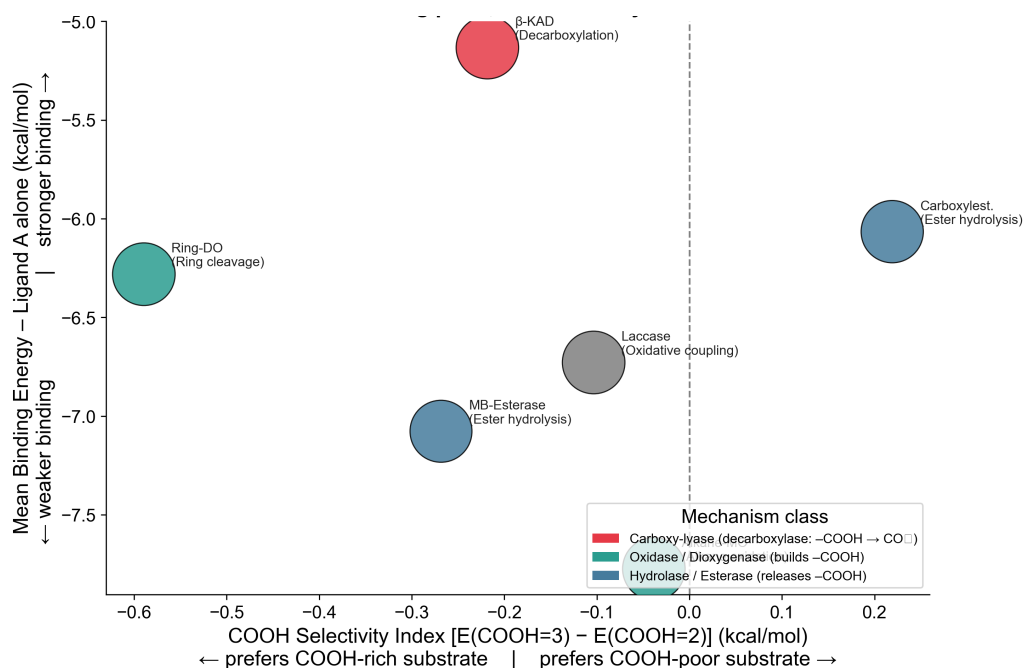
Supplementary Figure S14d. Binding free energy by marine environment.



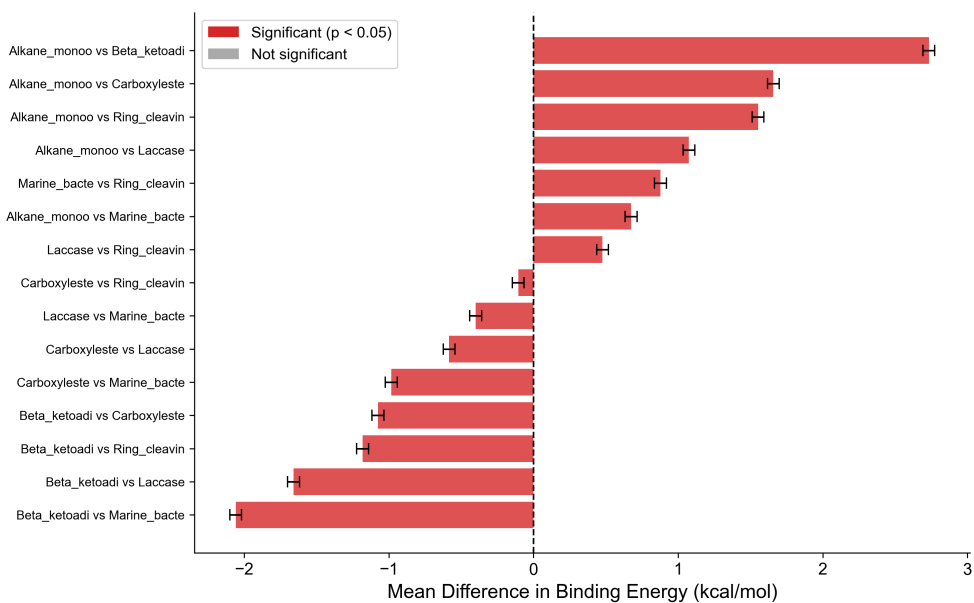
Supplementary Figure S14e. pH effect on binding free energy (pH 5, 7, 9).



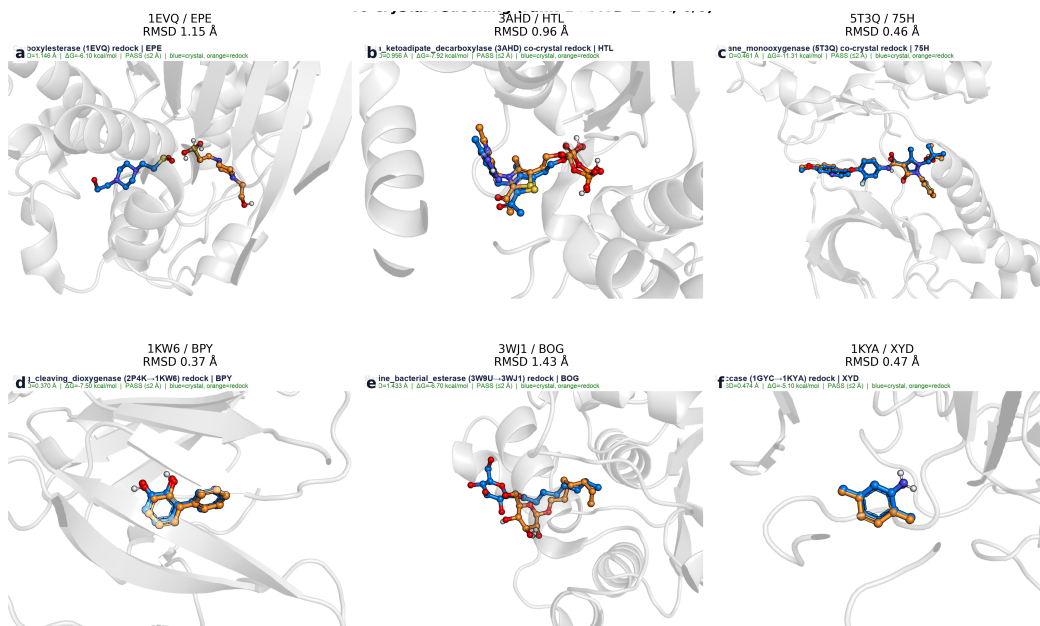
Supplementary Figure S14f. Binding free energy by nCOOH within each enzyme.



Supplementary Figure S14g. Carboxyl-count fate overview across enzymes (increase / decrease / unchanged).



Supplementary Figure S14h. Tukey HSD pairwise enzyme contrasts on binding free energy.

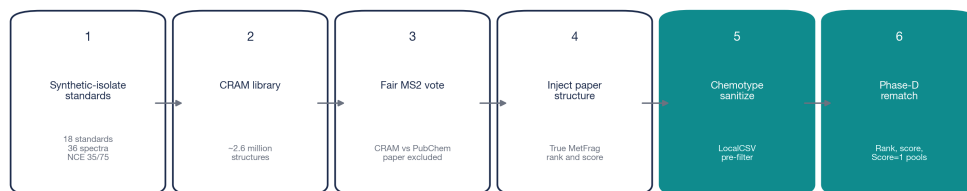


Supplementary Figure S14i. Co-crystal redocking: crystal (reference) versus rank-1 redocked pose.

Supplementary Note 10. Synthetic-standard MetFrag filter-sensitivity test

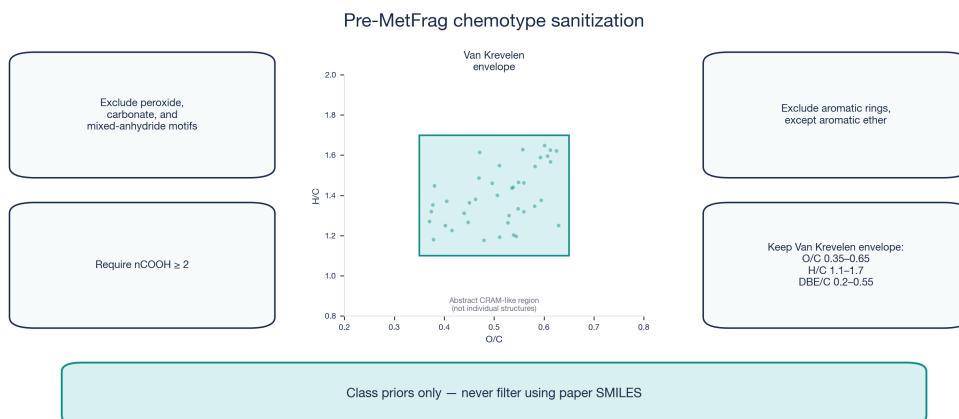
Supplementary Note 10. Synthetic-standard MetFrag filter-sensitivity test of the generated CRAM library. We asked whether class-prior chemotype pre-filtering of the MetFrag can shrink the candidate space and improve injected paper-standard rank without a material loss of MS² fidelity. Eighteen authentic synthetic-isolate (SI) standards (36 spectra at NCE 35 and 75) were redrawn from Craig et al., Environ. Sci. Technol. 2024 (DOI 10.1021/acs.est.4c00166); Hawkes et al., Anal. Chem. 2025 (DOI 10.1021/acs.analchem.5c02665); and Flygare et al., Environ. Sci. Technol. 2026 (DOI 10.1021/acs.est.6c01032). Exact InChIKey coverage in the generated library is 2/18, so an overlay inject was required; paper SMILES were never used as a filter. Results and Fig. S18 open with native exact InChIKey coverage 2/18 and paper-in-Score=1 on 0/36; identifiers of the two recoveries, whether either is among the 16 strict-CRAM training exemplars, and maximum Tanimoto to the 18 standards are not tabulated in the deposited analysis tables. Median injected paper rank 62.5 → 3.0 after Phase-D chemotype sanitization is a CRAM-definition prune / filter-sensitivity result, not library validation (mean rank 77.25 → 16.08; Rank-1 0/36 → 10/36; Top-5 2/36 → 27/36; Top-10 3/36 → 29/36; 32 improved, 0 worsened, 4 unchanged; mean candidates scored 401.22 → 303.78; paper FragmenterScore mean 0.75 → 0.86; top ExplPeaks cosine 0.853 → 0.834, Delta -0.02, not a non-inferiority claim). Score=1 ties occurred on 18/36 spectra (max pool 35) with always one unique ExplPeaks string among Score=1 candidates, and the paper structure was in the Score=1 pool on 0/36 spectra, so offline Score=1 filters cannot raise paper rank. Phase-D recipe: ban peroxide/carbonate/mixed-anhydride SMARTS; nCOOH >= 2; no aromatic except aromatic with ether; van Krevelen O/C 0.35–0.65, H/C 1.1–1.7, DBE/C 0.2–0.55. Formula cap: first 400 rows per formula.

Evaluation and chemotype-rescore workflow

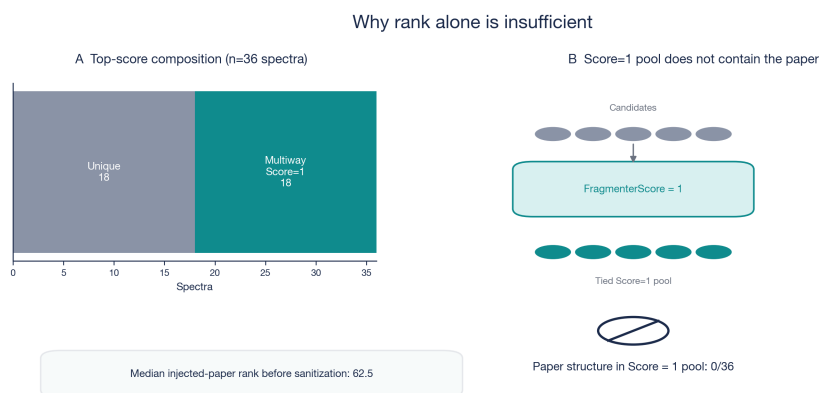


Paired before/after comparison for every spectrum

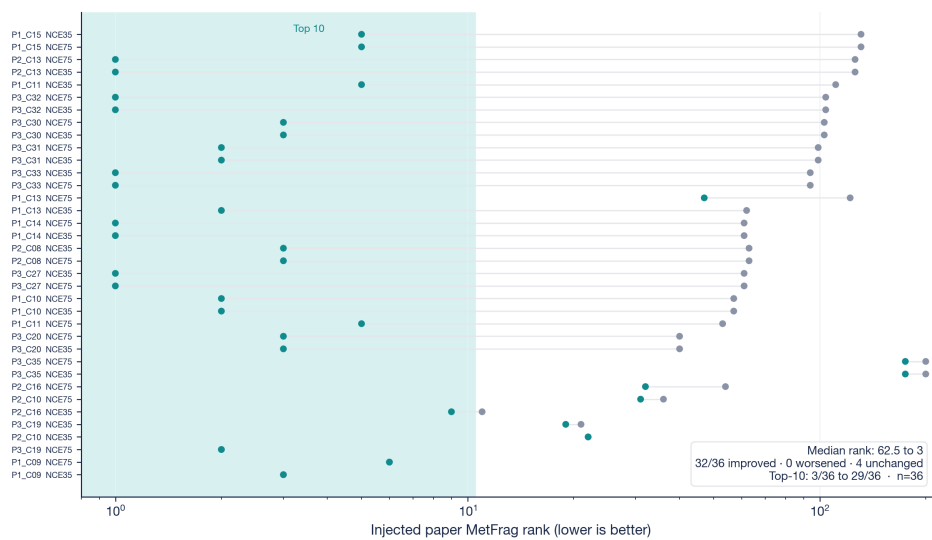
Supplementary Figure S15a. MetFrag filter-sensitivity workflow.



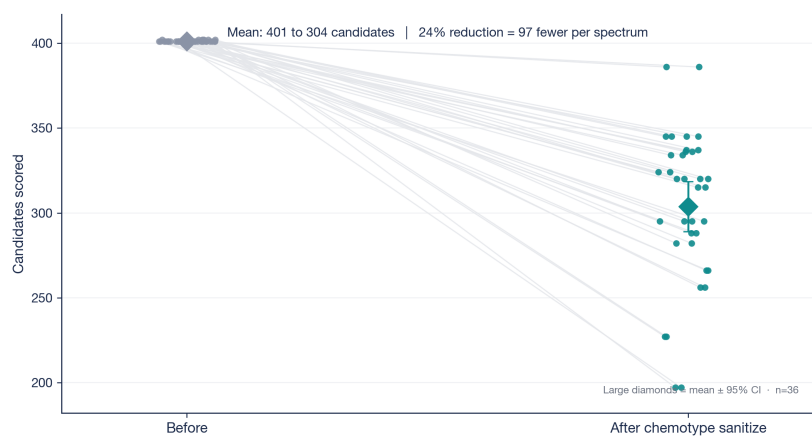
Supplementary Figure S15b. Chemotype filter recipe (Phase D).



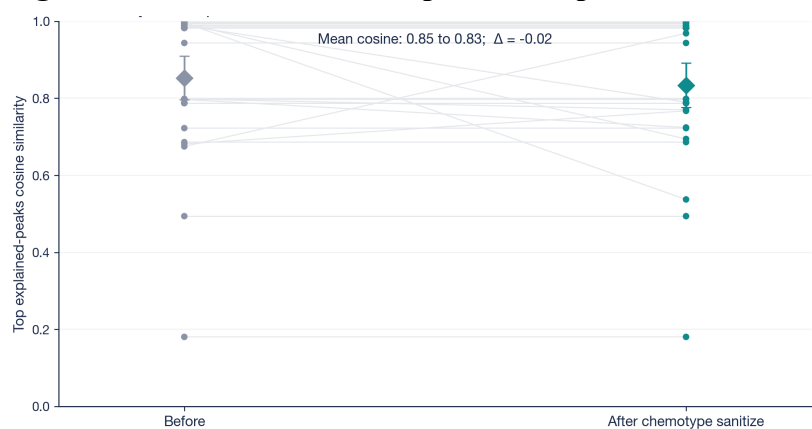
Supplementary Figure S15c. Why Score=1 rank alone is insufficient.



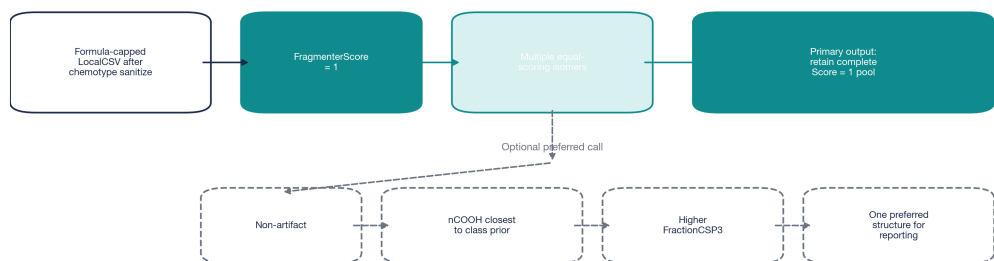
Supplementary Figure S15d. Injected paper-standard rank.



Supplementary Figure S15e. Candidate-count paired comparison.



Supplementary Figure S15f. MS2 ExplPeaks cosine near-preservation.



Supplementary Figure S15g. Same-score (Score=1) pool policy.

Supplementary Tables

Table S1. Network topology by environment (supports main Table 2).

Environment	Compounds	Density	Average degree	RXN_types	Mean molecular weight (Da)
Coastal	25955	0.0251	7.53	52	311
Deep Sea	18010	0.0162	4.86	55	339
Coral Reef	15453	0.0109	3.25	50	301
Open Ocean	13843	0.0161	4.82	53	314

River 6405 0.0415 12.4 50 280

Table S2. Formation-edge reaction-class means by environment (supports main Table 2 oxidation share and O/R).

Transformation	Coastal	Deep sea	Open ocean	Coral reef	River
Oxidation	28.4 ± 8.9	35.2 ± 7.6	32.4 ± 15	34.1 ± 5.8	29.4 ± 9.9
Alkylation	12.8 ± 6.2	11.2 ± 3.3	14.6 ± 8.8	11.9 ± 3.7	15.3 ± 8.1
Hydrolysis	12.1 ± 9.3	10.1 ± 3.7	12 ± 10.6	11 ± 1.6	10 ± 5.6
Reduction	10.3 ± 4.6	10.9 ± 2.8	10 ± 3.3	9.4 ± 2.3	10.7 ± 8.9
Acylation	4.4 ± 2.8	3.7 ± 1.4	4.4 ± 2.8	3.7 ± 0.4	5.3 ± 3.3
Carboxylation	3.7 ± 2.3	4.3 ± 1	4 ± 2.8	3.8 ± 1	4.6 ± 1.5
Glycosylation	2.6 ± 1.7	1.9 ± 1.3	2.7 ± 3.1	3.1 ± 0.7	2.7 ± 1.9

Table S3. NPClassifier pathway means ± SEM by environment.

Pathway	Coastal	Deep sea	Open ocean	Coral reef	River
Terpenoids	19.6 ± 1.27	25.6 ± 3.34	20.5 ± 1.95	19.6 ± 2.43	25.2 ± 5.07
Shikimates	20 ± 2	10.6 ± 1.22	12.9 ± 2.14	15.8 ± 2.8	15.8 ± 2.11
Alkaloids	24 ± 1.41	23.6 ± 1.8	27.7 ± 2.02	27.2 ± 2.84	18.6 ± 3.43
Amino acids	16.9 ± 1.27	22.3 ± 4.01	18.7 ± 1.67	21.5 ± 7.08	12.2 ± 2.29
Polyketides	12.5 ± 0.85	10.1 ± 0.84	11.6 ± 1.26	8.34 ± 1.56	21.7 ± 1.45
Carbohydrates	3.18 ± 0.46	1.69 ± 0.48	2.11 ± 0.33	2.83 ± 0.96	3.71 ± 1.37
Fatty acids	3.76 ± 0.59	6.13 ± 1.29	6.51 ± 1.17	4.75 ± 1.13	2.88 ± 1.09

Table S4. Oxidation/reduction percentages and O/R ratios (supports main Table 2 O/R).

Environment	Oxidation(%)	Reduction(%)	O/R ratio
Coastal	28.4	10.3	2.76
Deep Sea	35.2	10.9	3.23
Open Ocean	32.4	10.0	3.24
Coral Reef	34.1	9.4	3.63
River	29.4	10.7	2.75

Table S5. LC-MS/MS annotation corpus by environment (full database breakdown supporting main Table 2).

Environment	n	Spectra	PubChem	CRAM	COCONUT	ECMDB
Coastal	45	106051	30508	13364	6733	716
Deep Sea	11	42291	17177	7571	2330	482
Coral Reef	4	11527	5206	1776	721	239
Open Ocean	27	269259	84015	35576	14731	2434
River	11	47519	19088	6285	2913	616
Total	98	476647	155994	64572	27428	4487

Table S6. Docking competitive enzyme summary (extended metrics for main Table 3).

Enzyme	N Pairs	Mean E(A+B)	Mean Le(A)	Mean Deltadelta (G)	Cooperative (%)	Mean Lle(A)
Carboxylesterase	8892	-6.40 ± 0.73	0.299 ± 0.092	-0.08 ± 1.08	52.9%	-1.58

beta-Ketoadipate Decarboxylase	8964	-5.33 ± 0.77	0.247 ± 0.063	-0.14 ± 0.67	55.0%	-1.61
Alkane Monooxygenase	8988	-8.06 ± 0.98	0.372 ± 0.083	-0.11 ± 0.90	53.5%	-1.51
Ring-Cleaving Dioxygenase	8988	-6.51 ± 1.29	0.298 ± 0.071	-0.28 ± 0.96	57.9%	-1.58
Marine Bacterial Esterase	8802	-7.38 ± 1.03	0.342 ± 0.077	-0.08 ± 0.96	51.2%	-1.52
Laccase (control)	8988	-6.98 ± 0.82	0.324 ± 0.082	-0.06 ± 0.73	52.8%	-1.56

Table S7. Source-file matching rates (98 sources)

Environment	Matching rate (% mean ± SD)	n
Coastal	34.8 ± 11.3	45
Deep Sea	46.0 ± 11.9	11
Open Ocean	37.9 ± 17.3	27
Coral Reef	37.3 ± 6.1	4
River	33.4 ± 13.2	11

Table S8. CRAM annotation-consistency and end-product fractions. Network-topological metrics, not environmental residence time.

Environment	Annotation-consistency (%)	End-product (%)
Coastal	43.9 ± 15.8	22.4 ± 10.1
Deep Sea	49.3 ± 13.5	15.3 ± 10.1
Coral Reef	34.5 ± 23.7	27.6 ± 22.0
Open Ocean	42.8 ± 23.7	18.2 ± 11.6
River	44.7 ± 29.4	20.7 ± 16.3

Table S9. Selected CRAM-formation Markov transitions.

Transition	Environment	P (mean ± SD)
Oxidation -> Oxidation	Coastal	0.251 ± 0.276
Oxidation -> Oxidation	Deep Sea	0.386 ± 0.101
Oxidation -> Oxidation	Open Ocean	0.399 ± 0.278
Oxidation -> Oxidation	Coral Reef	0.272 ± 0.055
Oxidation -> Oxidation	River	0.196 ± 0.167
Oxidation -> Carboxylation	Coastal	0.014 ± 0.030
Oxidation -> Carboxylation	Deep Sea	0.022 ± 0.019
Oxidation -> Carboxylation	Open Ocean	0.023 ± 0.037
Oxidation -> Carboxylation	Coral Reef	0.015 ± 0.010
Oxidation -> Carboxylation	River	0.022 ± 0.024
Alkylation -> Hydrolysis	Coastal	0.023 ± 0.064
Alkylation -> Hydrolysis	Deep Sea	0.037 ± 0.033
Alkylation -> Hydrolysis	Open Ocean	0.042 ± 0.082
Alkylation -> Hydrolysis	Coral Reef	0.055 ± 0.037
Alkylation -> Hydrolysis	River	0.099 ± 0.177

Table S10. Individual-pose docking means by chemotype.

Chemotype	n	Mean DeltaG
benzene	35995	-6.84 ± 1.36
alicyclic	24814	-6.56 ± 1.24
aliphatic	23426	-5.72 ± 1.11

Table S11. Individual-pose docking means by nCOOH.

nCOOH	n	Mean DeltaG
1	43461	-6.20 ± 1.43
2	32775	-6.68 ± 1.19
3	7193	-6.86 ± 1.18
4	806	-6.61 ± 1.21

Global RXNMapper confidence**Table S12A. RXNMapper confidence and similarity for mass-difference edges.**

Reaction class	n edges	Mean conf	Mean sim
Oxidation	467284	0.542±0.144	0.322±0.215
Alkylation	251919	0.566±0.158	0.363±0.241
Hydrolysis	151434	0.539±0.145	0.328±0.218
Reduction	149889	0.546±0.147	0.332±0.221
Dealkylation	137899	0.573±0.162	0.373±0.243
Acylation	63569	0.586±0.162	0.422±0.233
Carboxylation	57544	0.557±0.150	0.353±0.218
Decarboxylation	57544	0.557±0.150	0.353±0.218
Condensation	50478	0.539±0.145	0.328±0.218
Deacylation	36403	0.581±0.160	0.405±0.232

Table S12B. RXNMapper confidence and similarity for mass-difference edges.

Metric	Coastal	Deep Sea	Open Ocean	Coral Reef	River
All_reactions_Conf	0.520±0.045	0.584±0.037	0.562±0.054	0.616±0.019	0.577±0.081
All_reactions_Sim	0.301±0.084	0.391±0.055	0.354±0.075	0.423±0.036	0.338±0.134

Table S12C. CRAM vs all-network transformation enrichment.

Transformation	Enrichment	CRAM %	All-net %
Hydroxylation	1.69x	10.0	5.9
Formylation	1.65x	7.4	4.5
Carboxylation	1.61x	5.0	3.1
Deoxygenation	1.64x	10.0	6.1
Decarboxylation	1.57x	4.8	3.0
Acetylation	1.86x	7.5	4.0
Deacetylation	1.91x	7.6	4.0
Ketone formation	>50x	4.9	0.1
Ring opening	>50x	2.9	0.1
Cyclization	>50x	2.5	0.1

Transformation	Enrichment	CRAM %	All-net %
Methylation	0.59x	7.9	13.4
Ethylation	0.43x	4.0	9.4

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